Materials for Tissue Engineering BM-612

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The Graduate (1967)

For others what is pollution, for us that is solution

Mission Mangal (2019)

Content:

- Polymers as biomaterials for different applications
- Artificial polymers for biomaterials
- Biopolymers
- Modifications of Biopolymers
- Polymers and crosslinking
- Swelling of polymers
- Biodegradation
- Hydrogels
- Self healing polymers
- Shape memory polymers
- Polymers for surface modifications
- Polymers and drug delivery
- Polymers for actuators

Polymers as Biomaterials:

- Polymers for permanent application (non-biodegradable synthetic polymers) : Polyethylene, Polyacrylics, Poly Vinyl Chloride

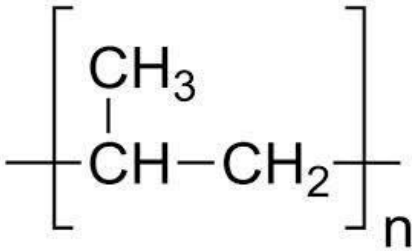
- Polymers for temporary structural application (biodegradable synthetic polymers): Poly caprolactone, Poly lactic acid

- Polymers from biological origins…biopolymers: Chitosan, Alginate, Collagen

- Polymers for nanomedicine and drug delivery: Polyethylene glycol, Poly (lactic-co-glycolic)acid, Pullulan

Nature of Repeat Units

Homopolymer- A-A-A-A-A-A-A-A-A-A-A-A-
(identical repeat units)



Polypropylene

Copolymer - Different repeat units (A &B)

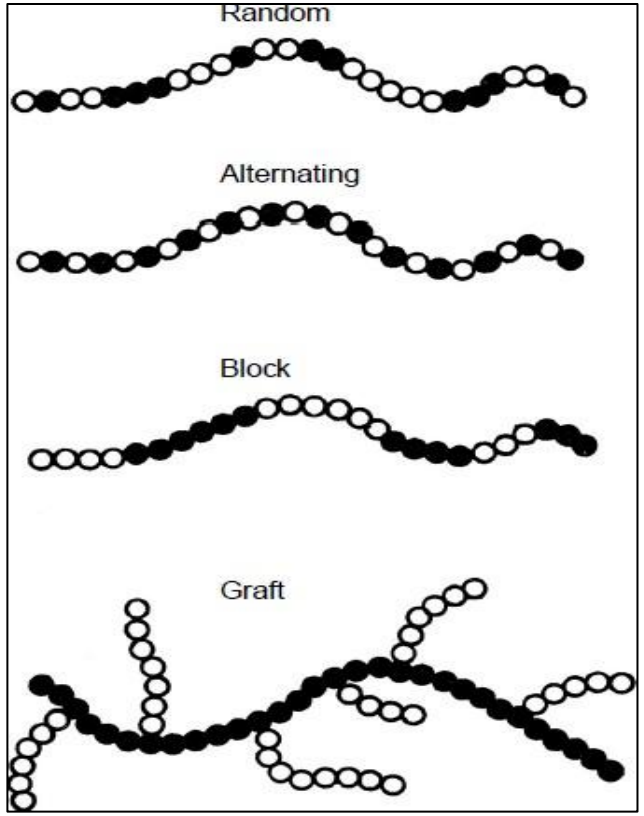
Alternating Copolymer- A-B-A-B-A-B-A-B-A-B

Block Copolymer - AAAAAAAA-BBBBBBBBB

Random Copolymer- AA-B-A-B-B-A-A-A-B-B-A

Graft Copolymer-

```
-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-A-
          -B                  -B
          -B                  -B
          -B                  -B
          -B                  -B
          -B                  -B
          -B                  -B
```



Molecular weight of Polymers

Equations for number average and weight average molecular weights as well as polydispersity index are defined as follows:

$$\bar{M}_n = \frac{\sum n_i M_i}{\sum n_i} = \frac{\sum w_i}{\sum n_i}$$

$$\bar{M}_w = \frac{\sum w_i M_i}{\sum w_i} = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

$$PD = \frac{\bar{M}_w}{\bar{M}_n}$$

where

M_i	molecular weight of molecules i
n_i	number of molecules with molecular weight i
w_i	mass of the molecules with molecular weight i

Mn (number avg molecular weight): measured by osmotic pressure
Mw (weight avg molecular weight): measured by dynamic light scattering

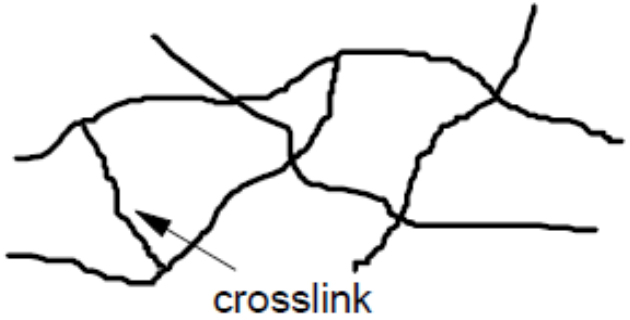
Structural characteristics-closely related to materials properties

linear (uninterrupted straight chain)

Poly-ethylene, PVC, Poly-amides

branched (occasional branches off longer chain)

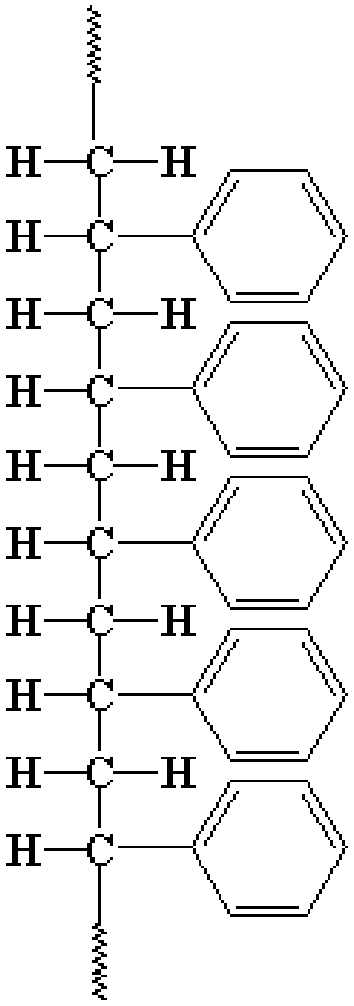
Low density poly-ethylene (LDPE)



networked (many interconnected linear chains; one giant molecule)

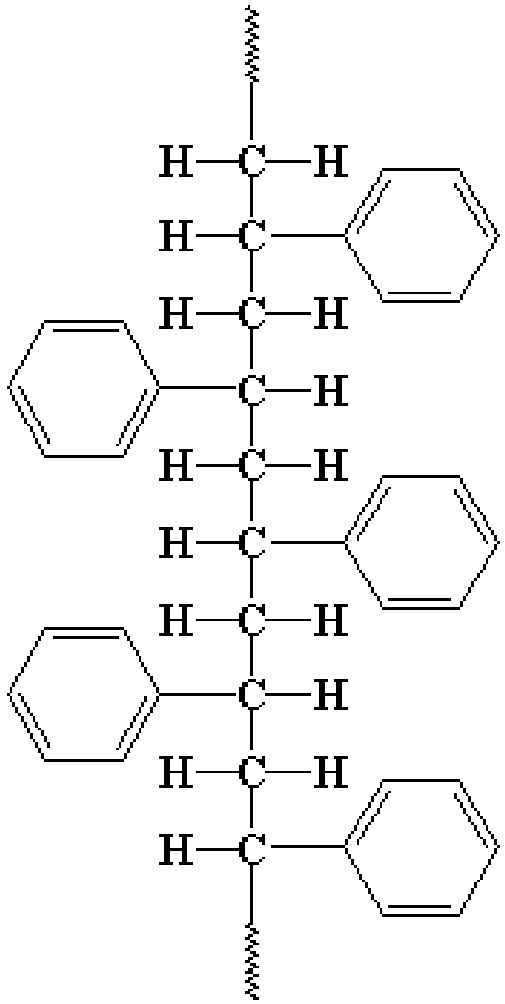
Epoxies, Rubber

Stereochemistry of side pendant groups



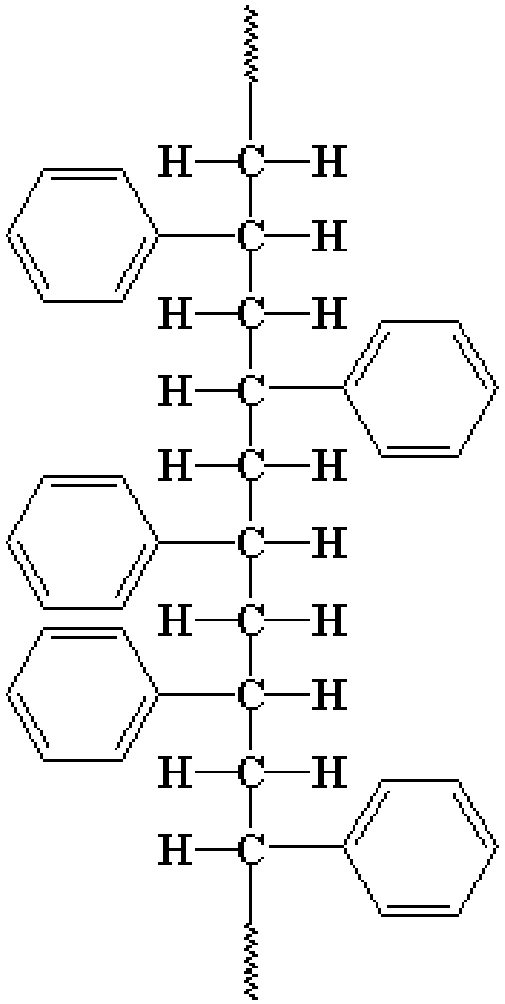
Same side

Isotactic



Alternating side

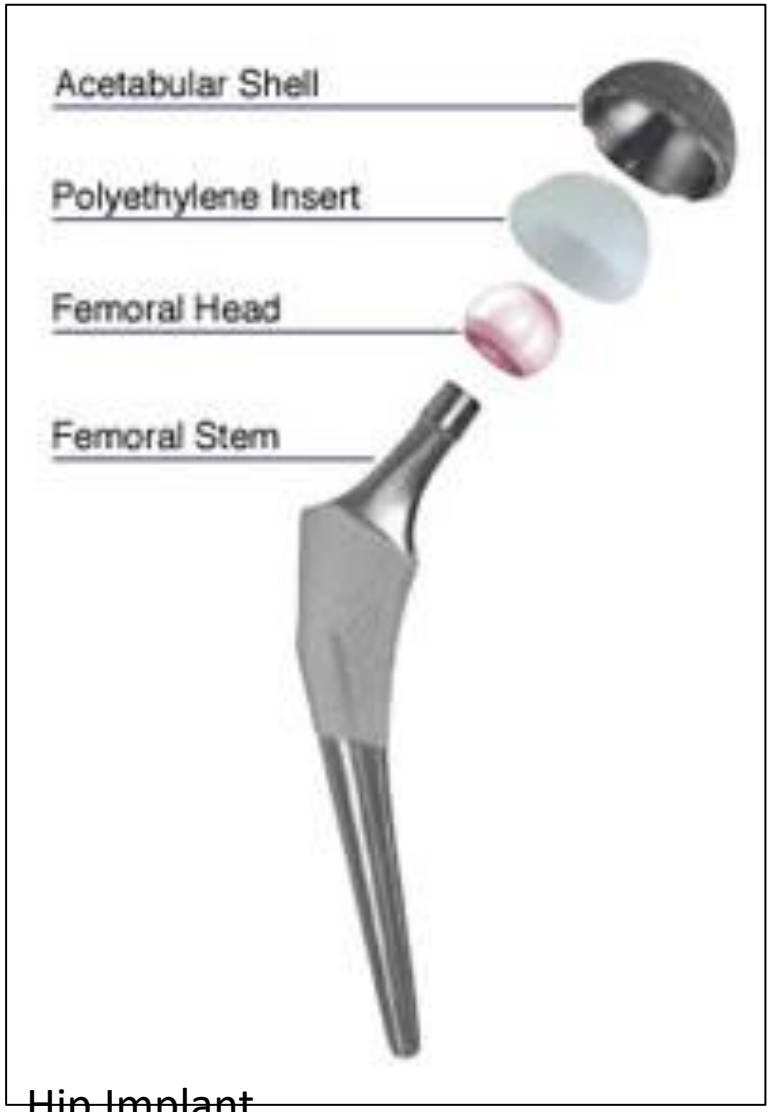
Syndiotactic



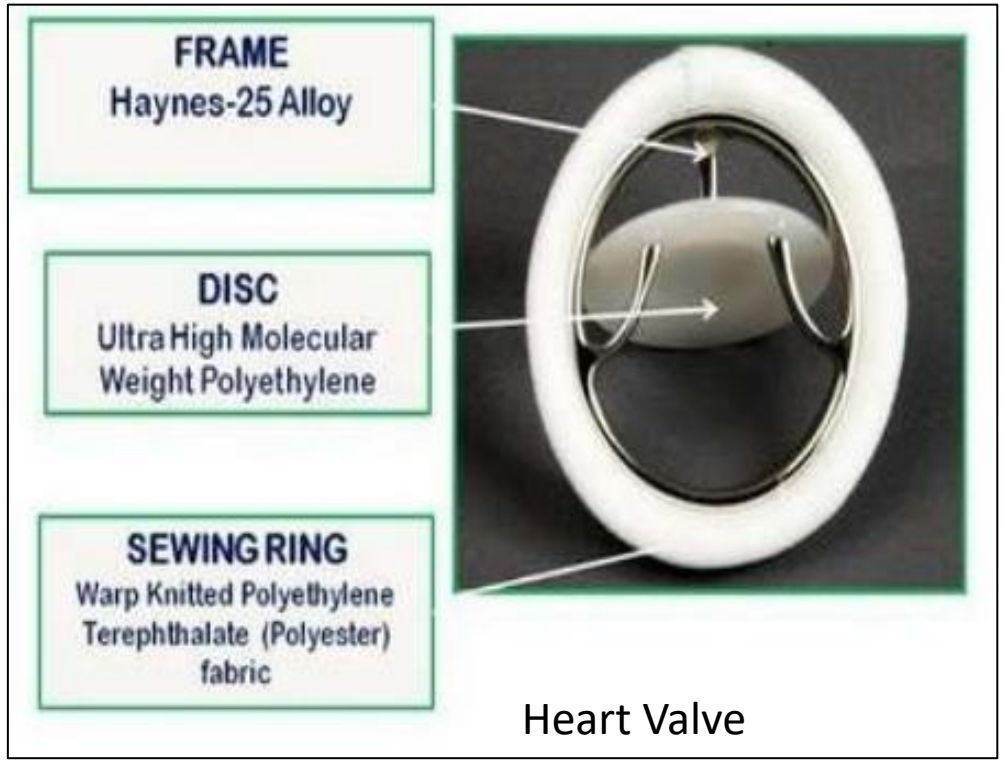
Random

Atactic

Bioinert polymers for biomaterial applications

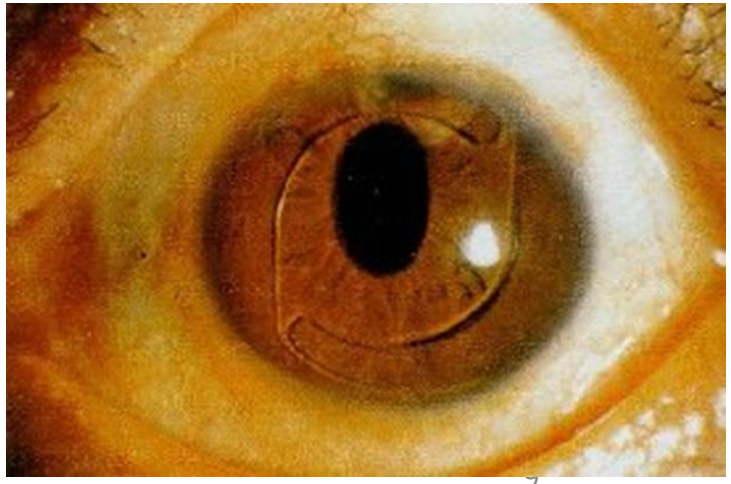


Hip Implant



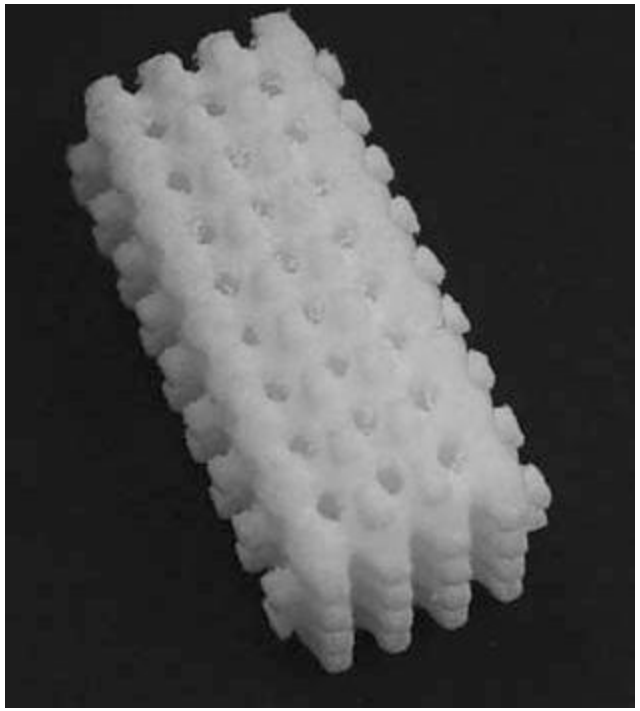
Heart Valve

PVC blood bags

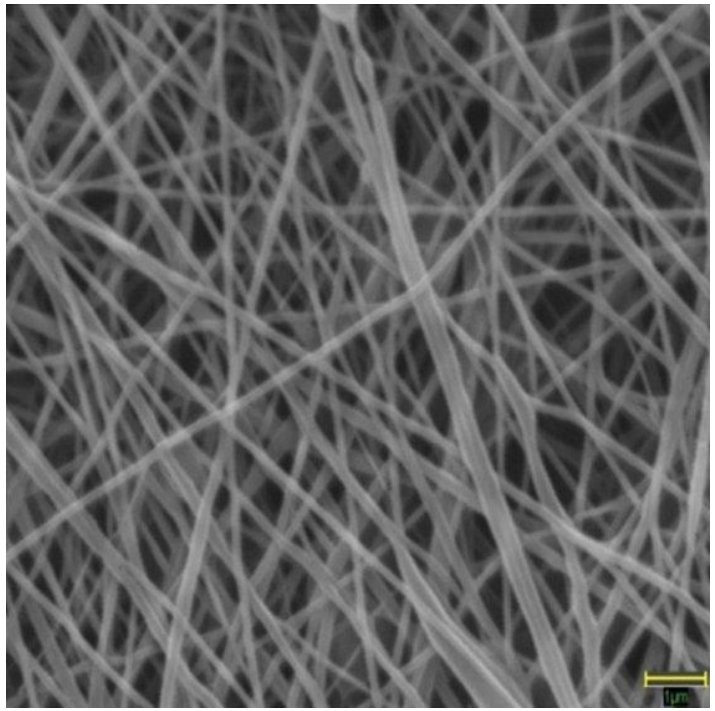


PMMA Intra-ocular lens

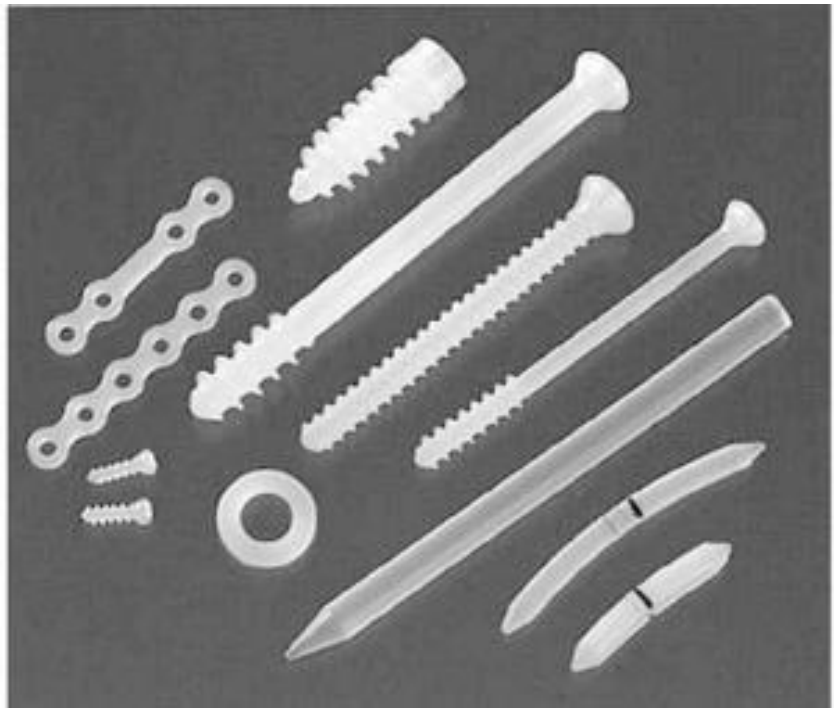
Biodegradable synthetic polymers



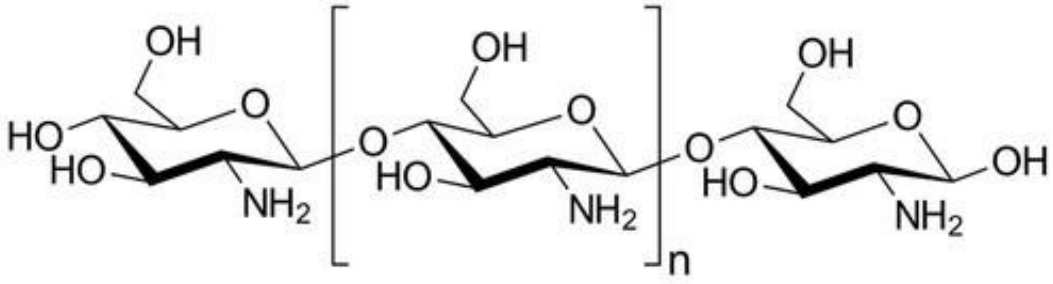
PCL 3D printed bone graft



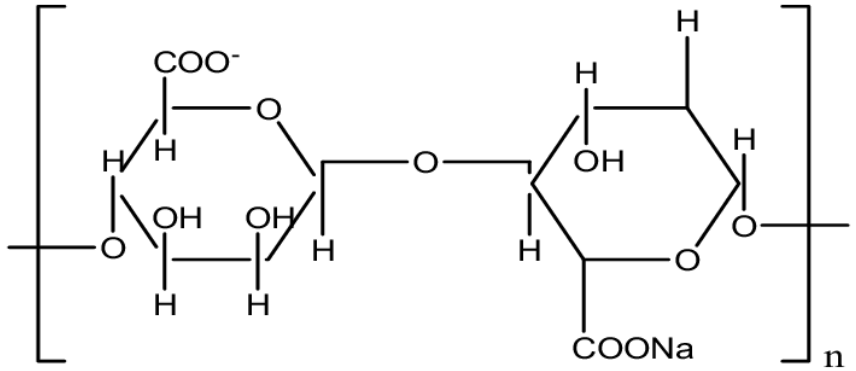
PCL nanofibrous mat for skin graft



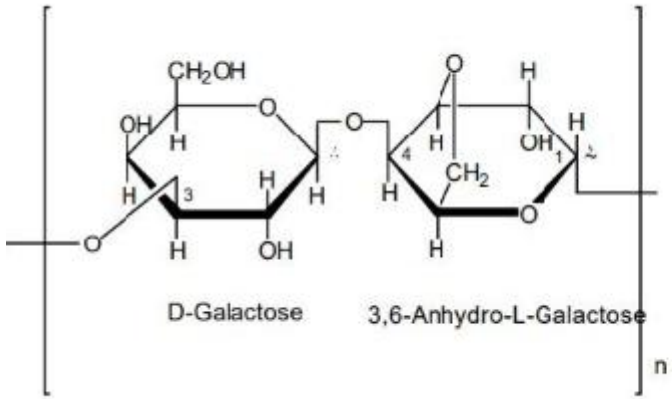
PLA bone plates, screws and nails

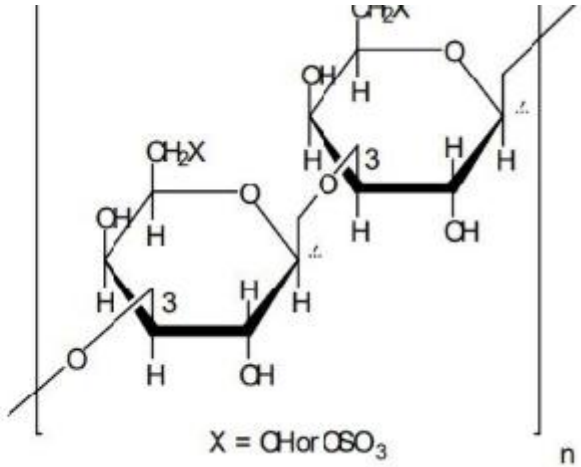


Chitosan: Crustacean shells

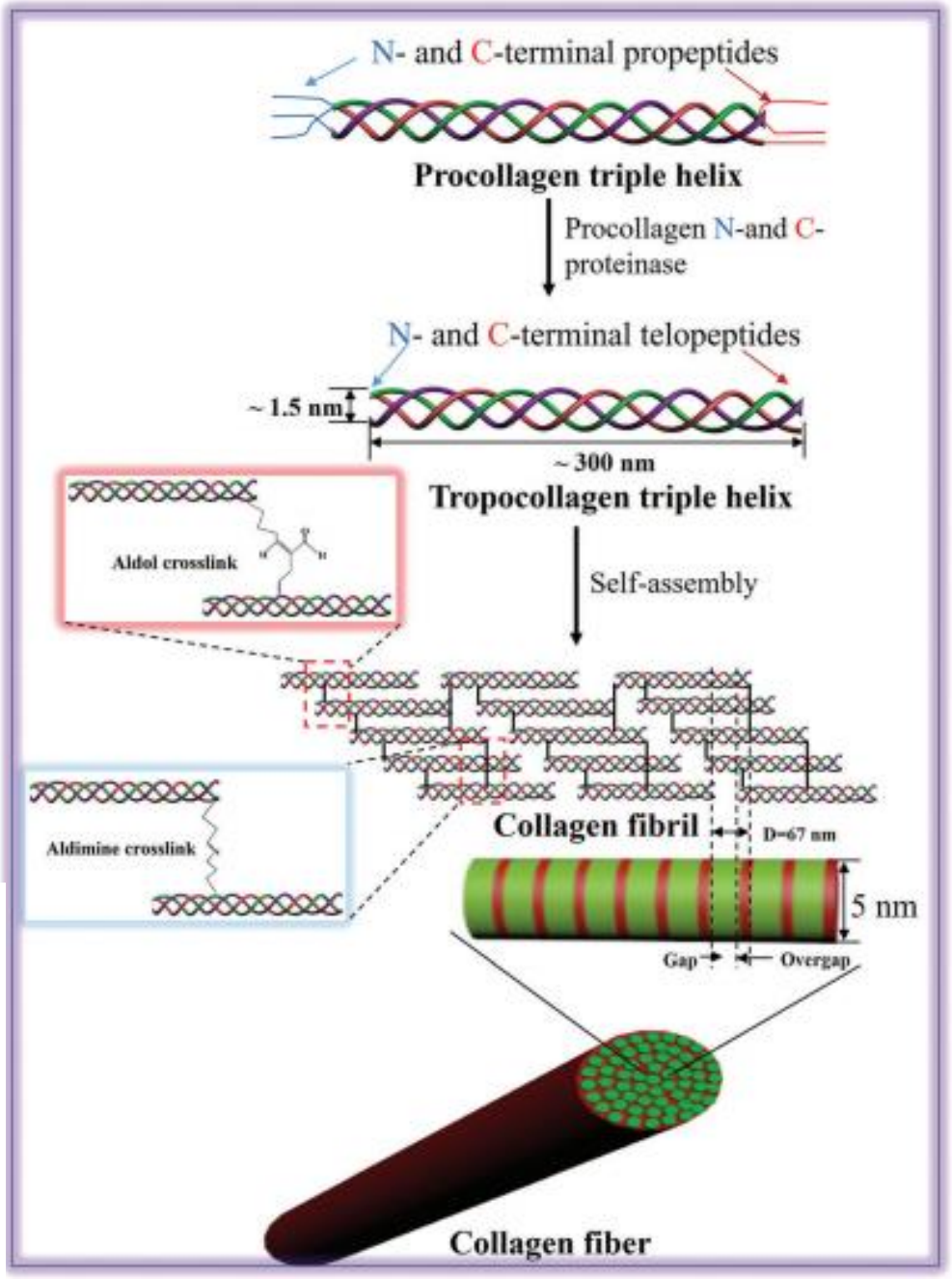


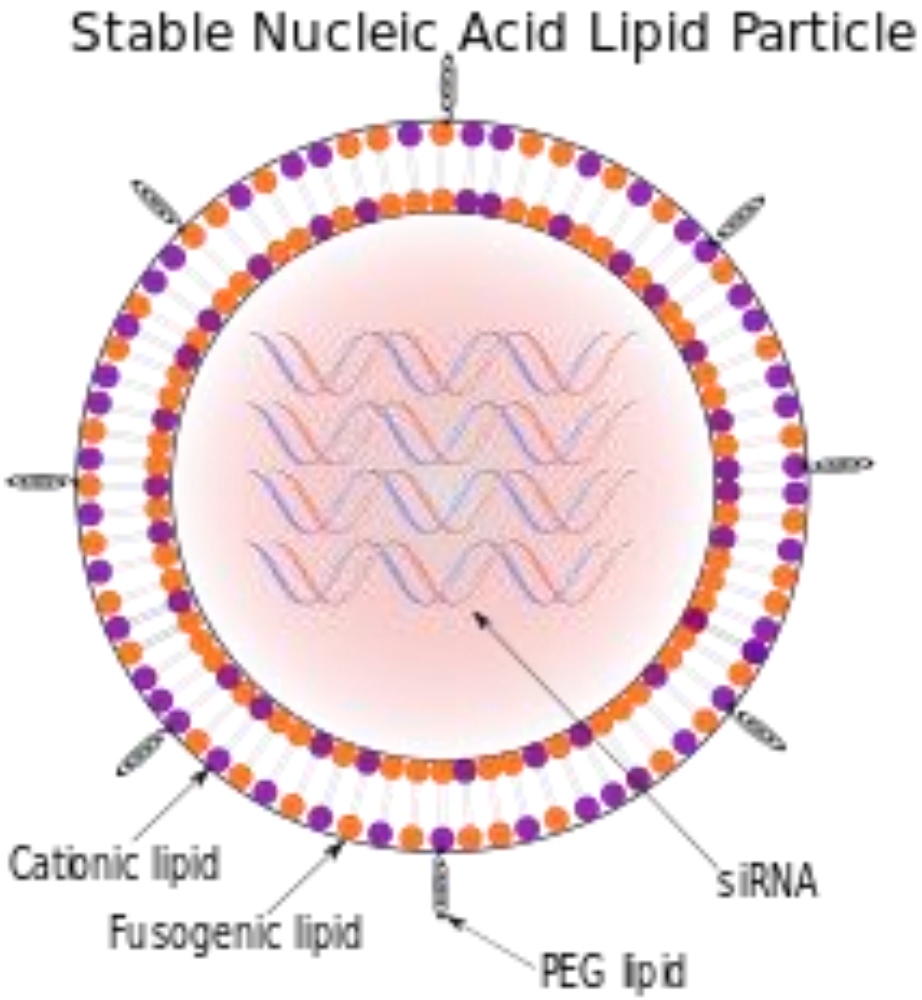
Alginate: brown seaweed

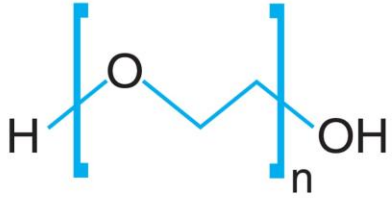




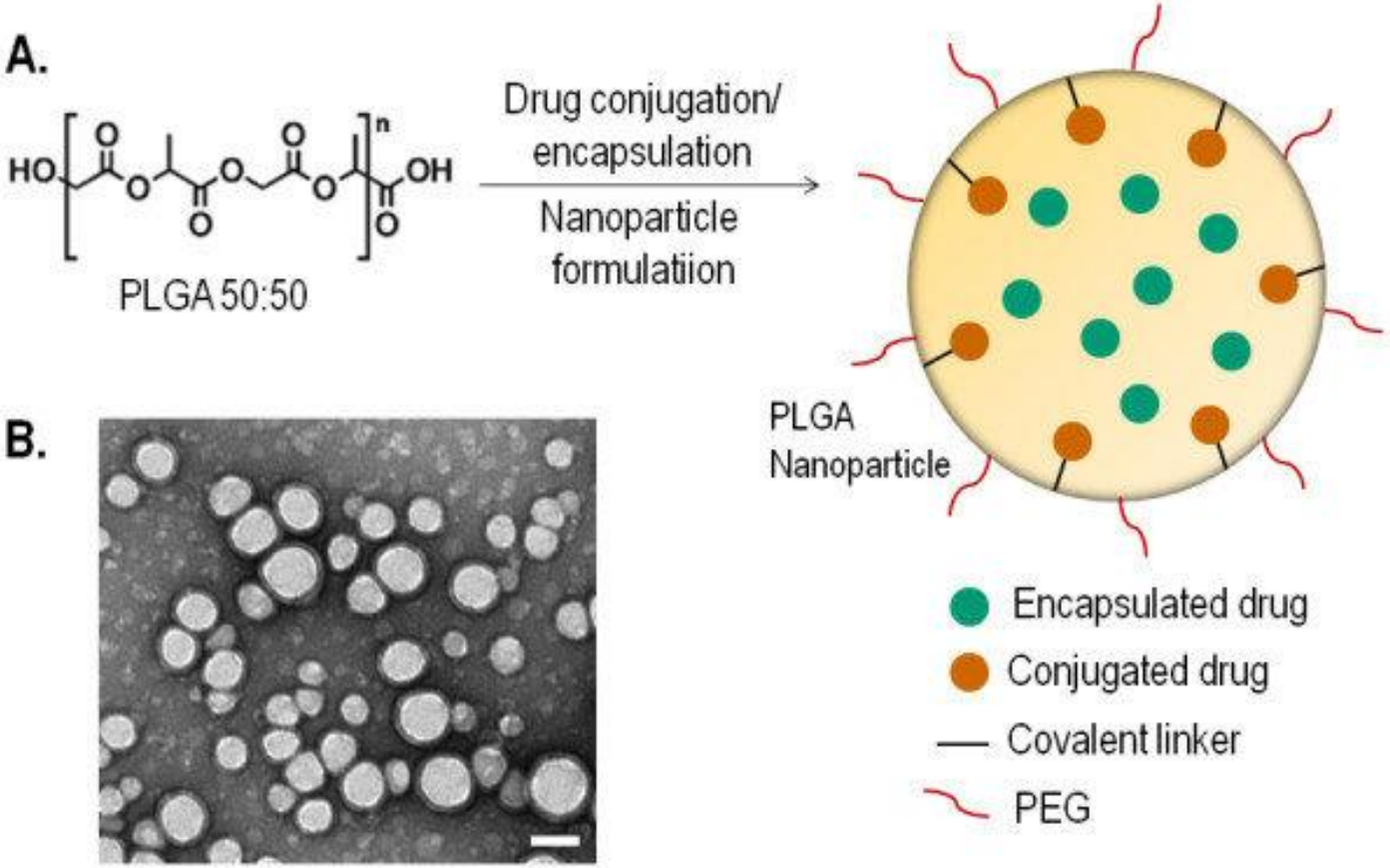
Agar



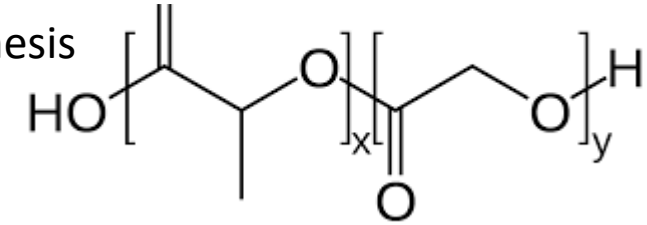




Lipid nanoparticles



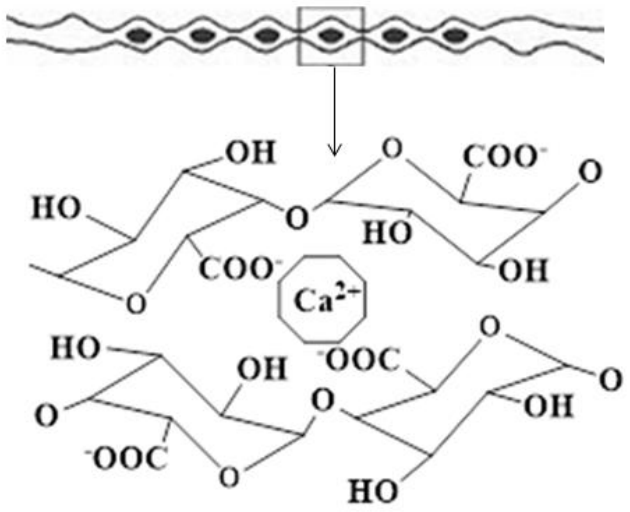
PLGA nanoparticle synthesis



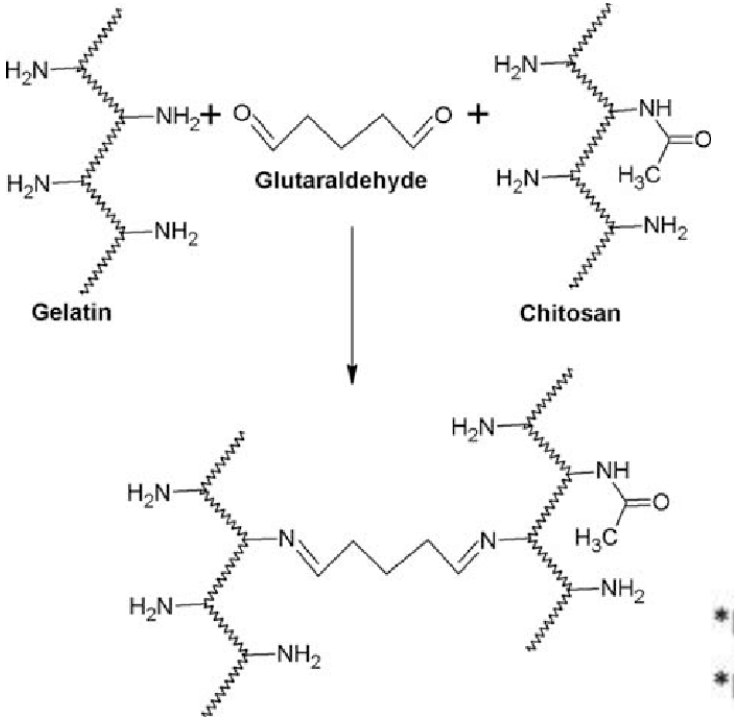
PEG functionalized solid lipid nanoparticles and PLGA nanoparticles for drug delivery

Cross linking of Polymer : Formation of inter or intrachain linkages in between polymer chains

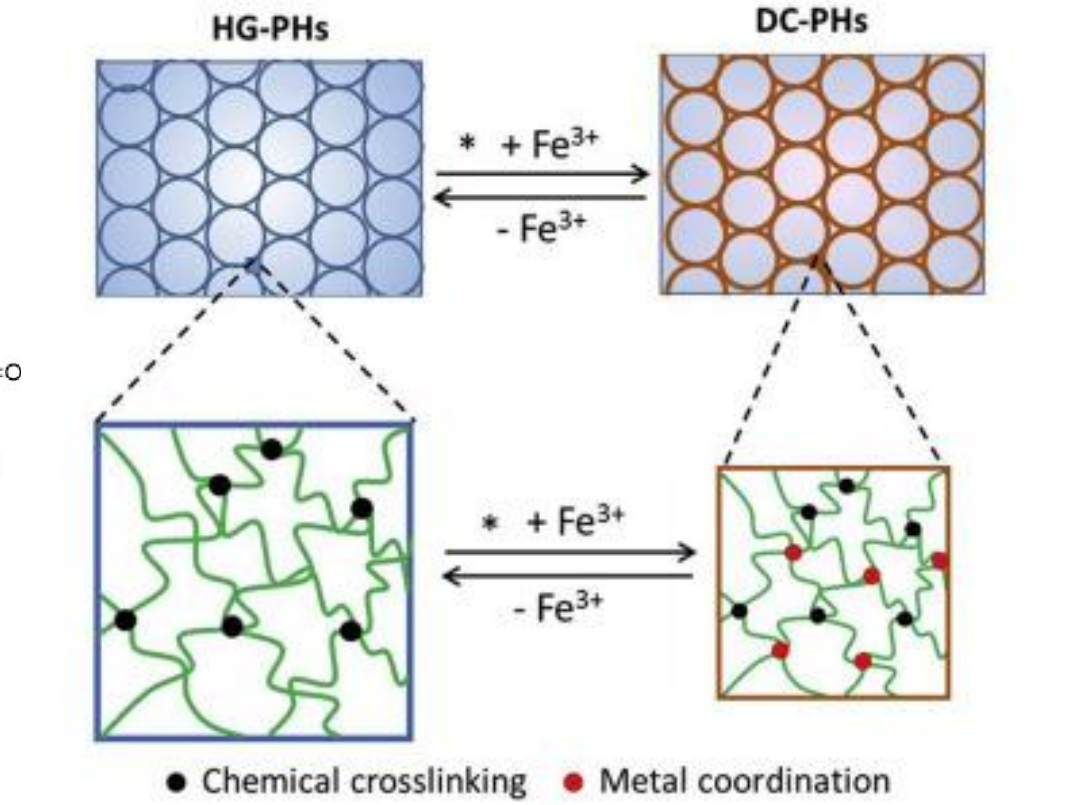
- Enhances mechanical strength
- Enhances thermal stability
- Reduces biodegradability
- Reduces swelling



Ionic crosslinking of alginate-Ca2+



Gelatin Chitosan crosslinking by Glutaraldehyde

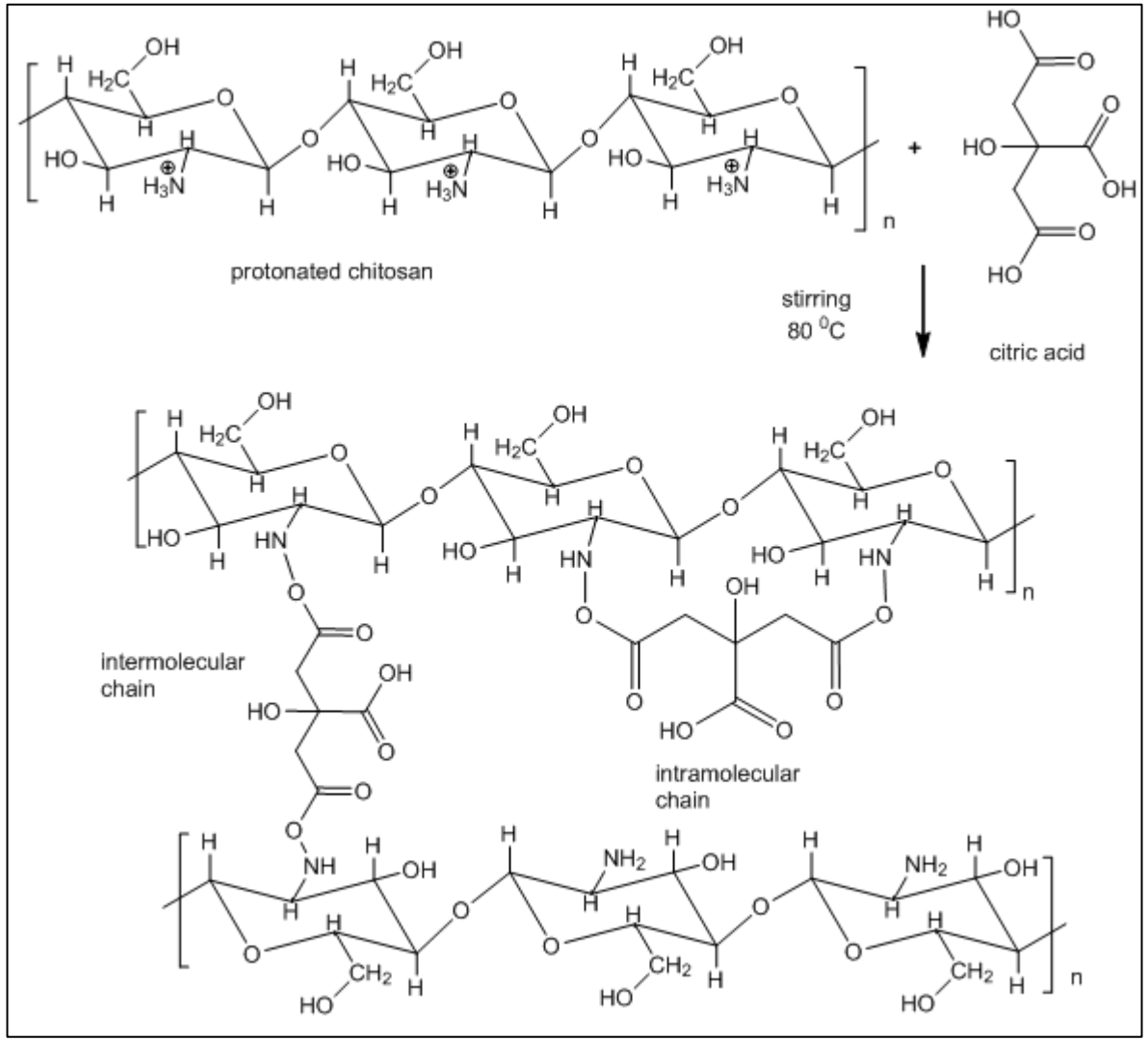


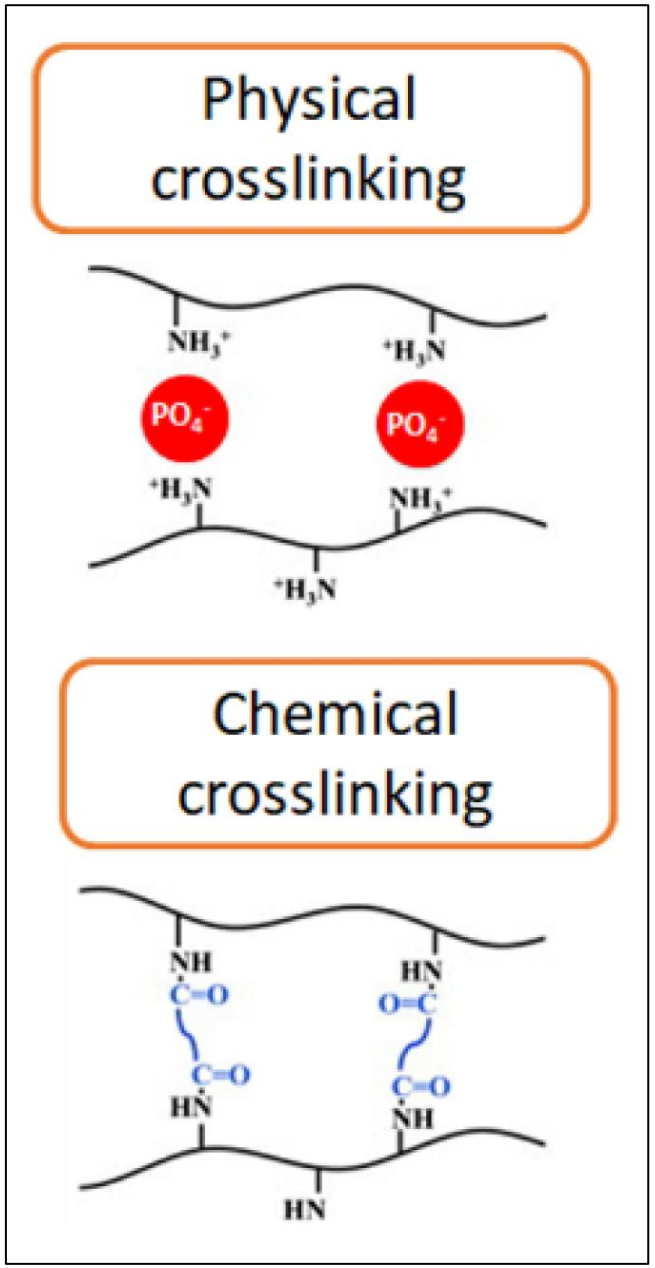
*Macroporous structures and rapid water absorption preserved.

*Mechanical properties enhanced and shape memory enabled.

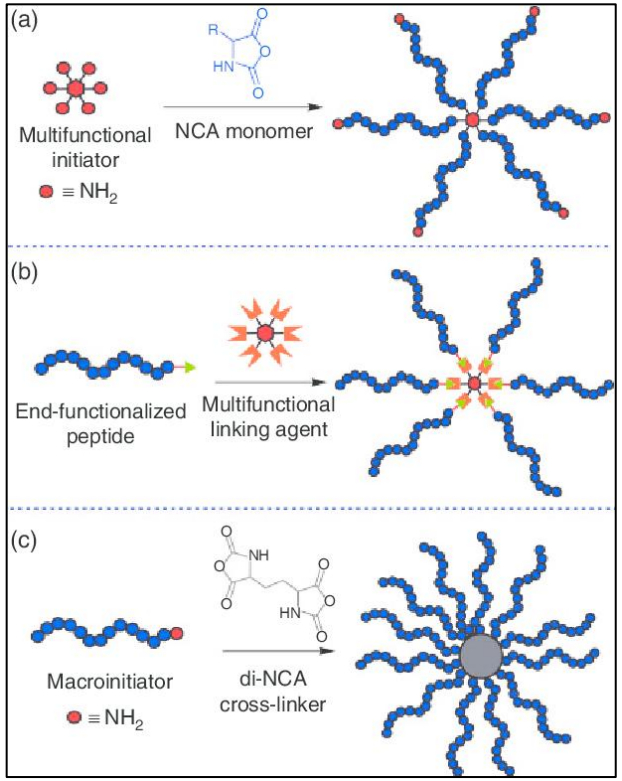
Metal coordination cross linking

Different crosslinking of same polymer:

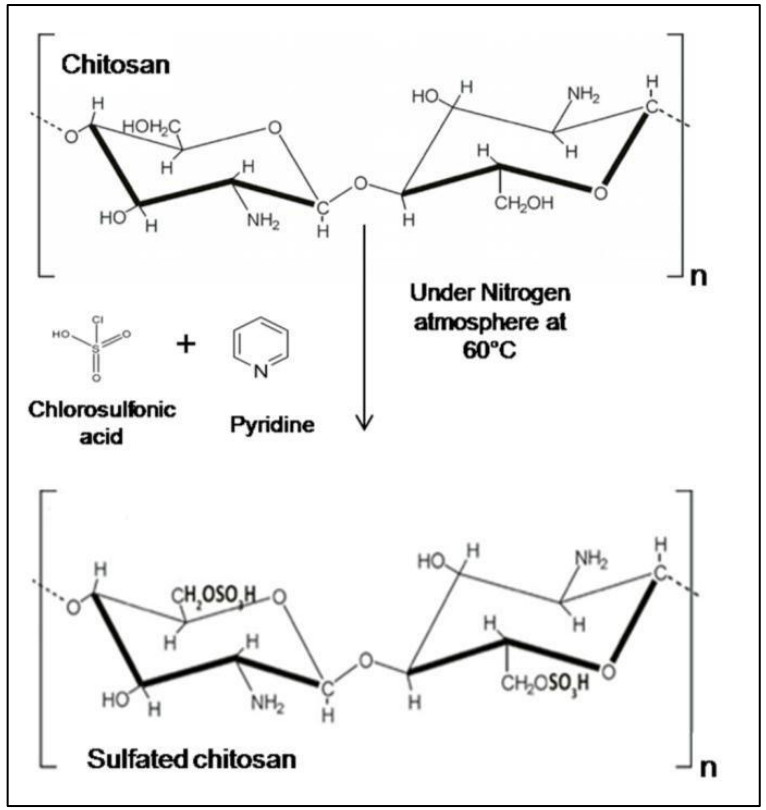




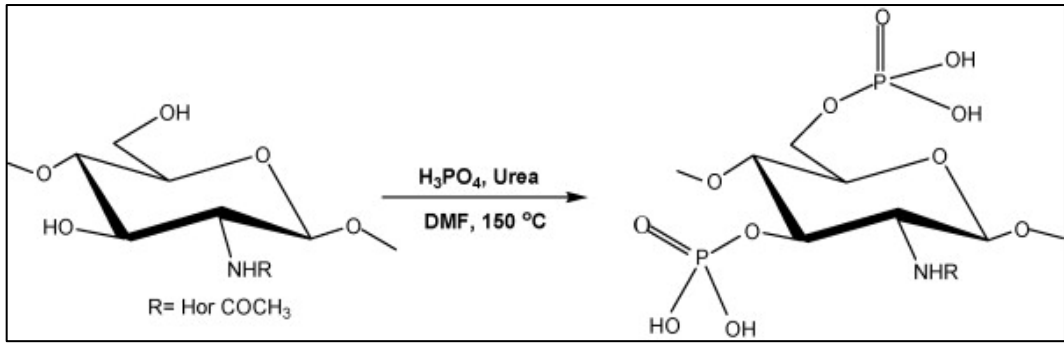
Enhancement of biocompatibility/bioactivity by chemical modifications of biopolymers:



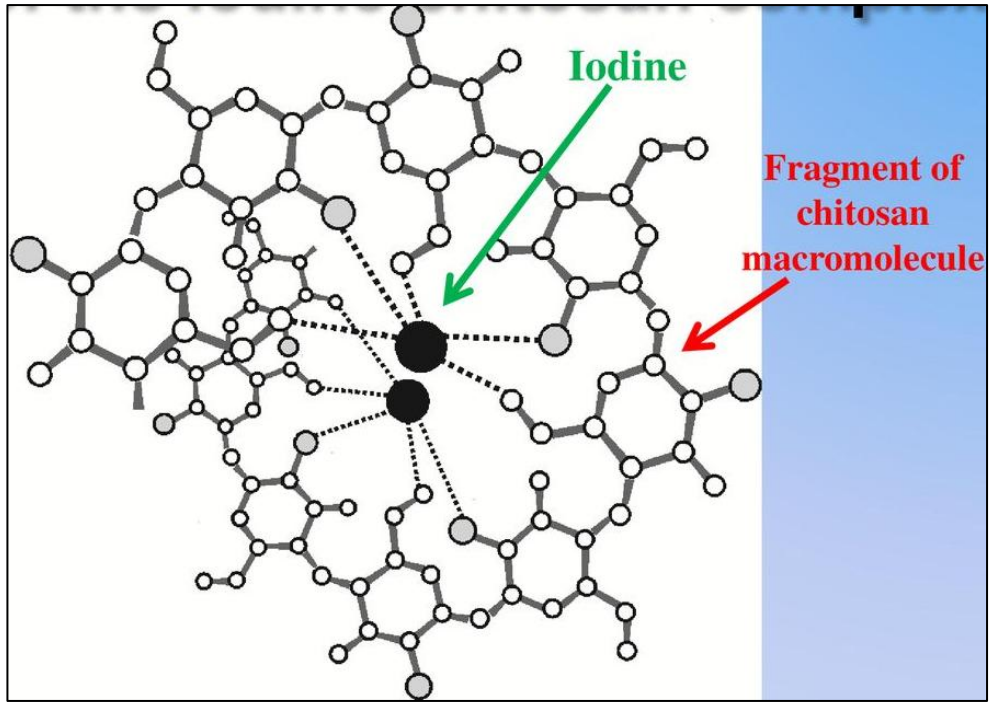
Peptide grafted polymer



GAGs mimetic polymer from chitosan



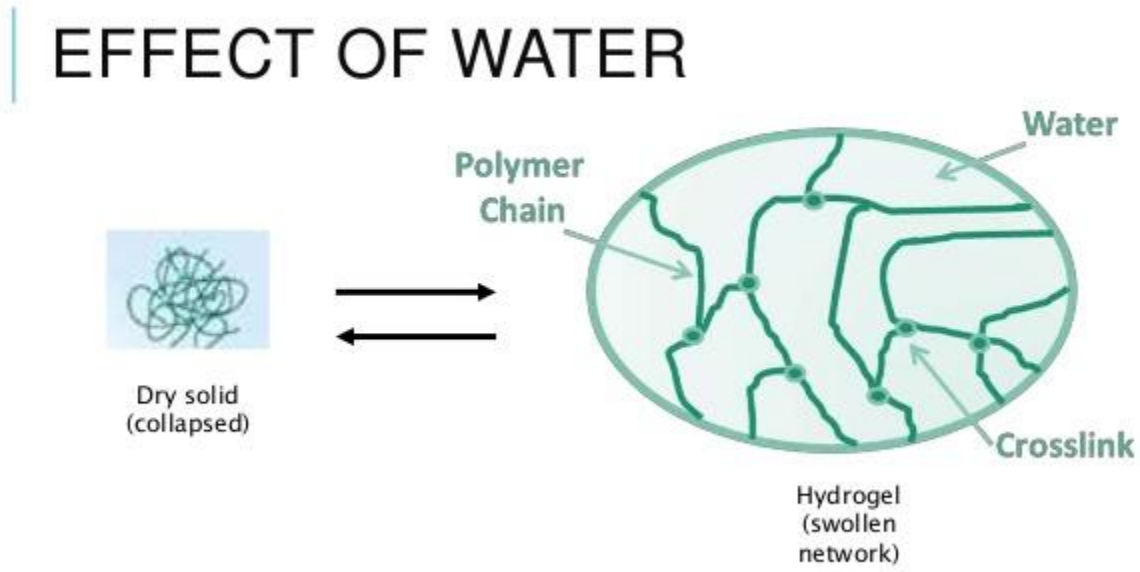
Phosphorylated chitosan for apatite like nucleation



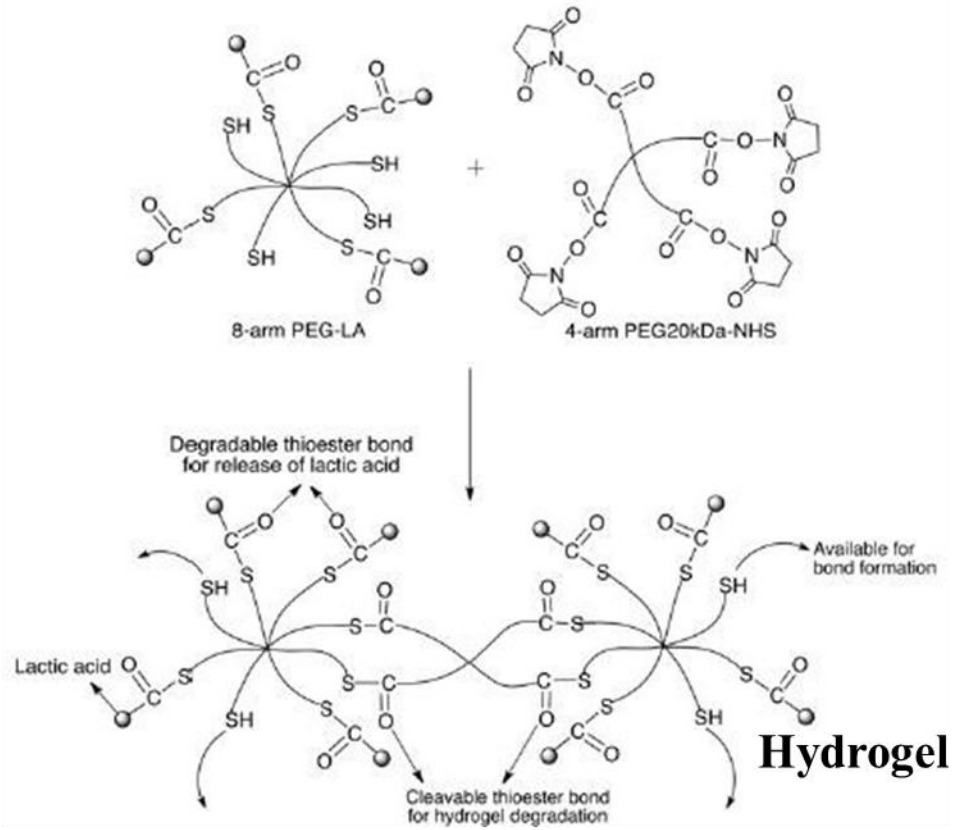
Iodinated chitosan for radio-opaque biopolymer

Swelling of Polymer: Every polymer has a tendency to absorb fluid (mostly water) up to a limit. During this process, the
volume of the polymer changes. The design of biomaterial always should consider this property to save the device from structural failure. Porosity and biomolecule release/diffusion kinetics is related to swelling Of polymer. The thermodynamic swelling force is counterbalanced by the retractive forces of the Crosslinked structure. Equilibrium is attained when the two forces become equal.
The volume of degree of swelling Q,
which is a ratio of actual volume of a sample in the swollen state divided by its volume in dry state, is then determined

Swelling ratio/capacity= [(w2-w1)/w1]*100
W1=initial weight of polymer before swelling
W2=weight of polymer post saturation with fluid



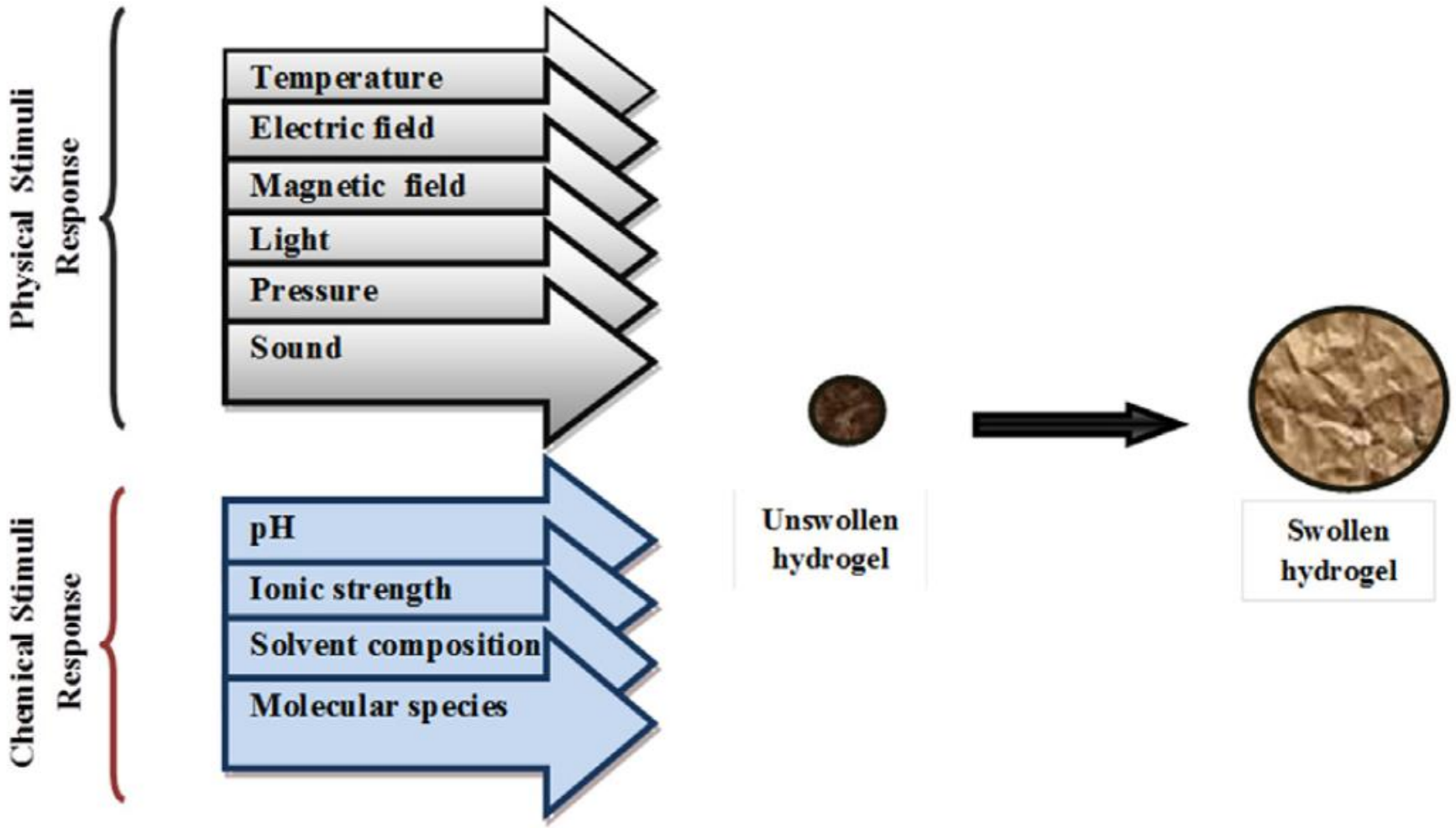
Hydrogels/gels



Hydrogel J Controlled Release 2014, 194, 301-309

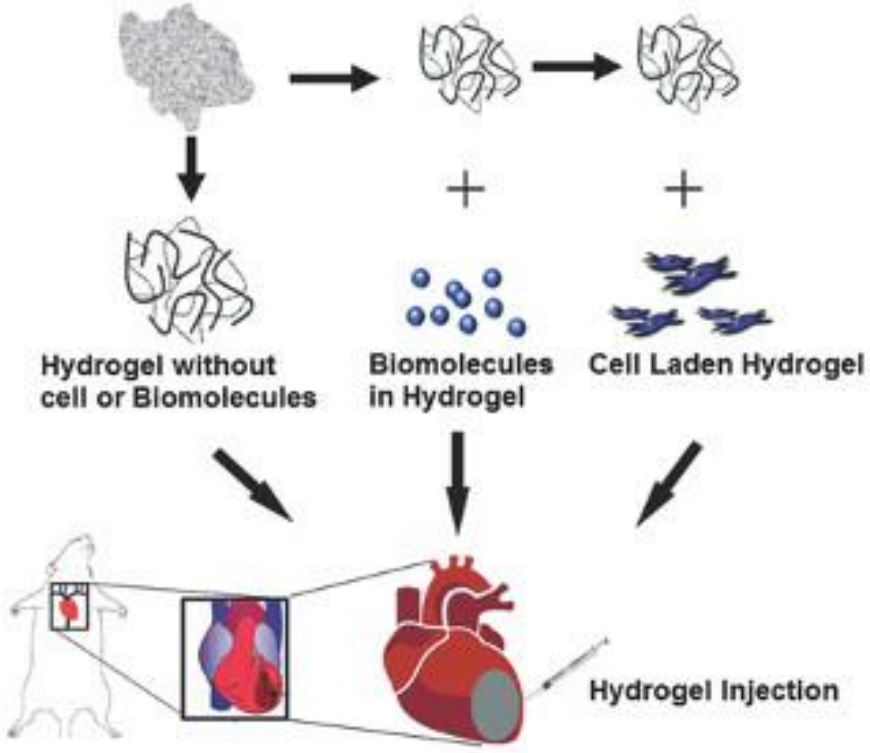
- Networks formed by covalent bonds are called **hydrogels**, whereas networks formed by non-covalent interactions are called **gels**
- The hydrogels/gels once formed, cease to flow.

Hydrogel sensitive to external stimuli



The widespread acceptance of hydrogels is primarily due to

- The structural similarity that they exhibit to macromolecular-based components in the body
- Their ability to replicate the properties of natural tissue- high water contents at equilibrium, exhibit rubbery behavior, and show low surface coefficients of friction
- Allow transportation of molecules (~primarily gases) needed for tissue function and survival
- Possess appropriate mechanical properties for contact with soft tissue

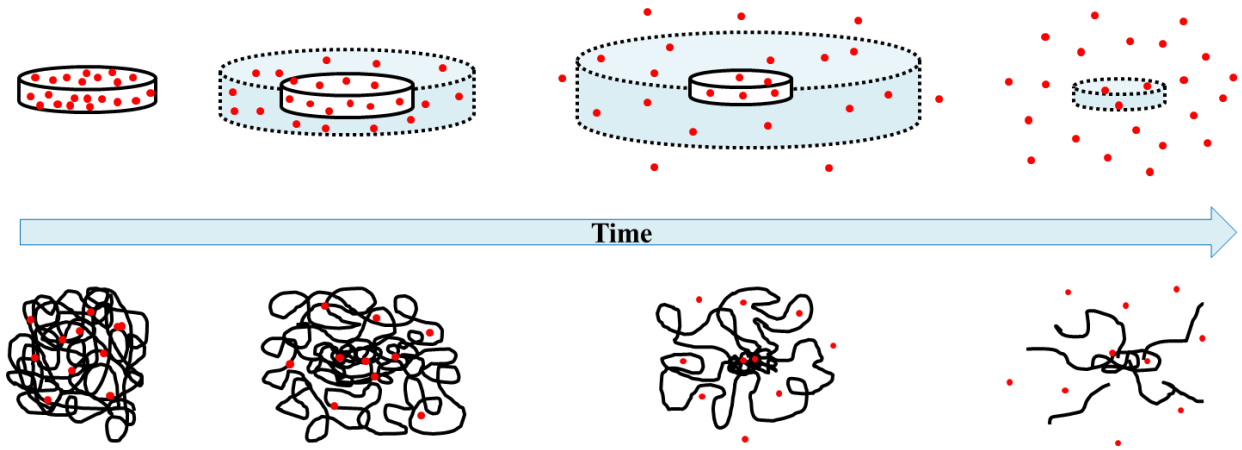


Hydrogels for tissue engineering

Synthetic hydrogel : (i) readily formed into highly hydrated materials (ii) physical and chemical properties are amenable to interaction with cells and tissues.

Material	Abbreviation	Application
Poly(ethylene glycol)	PEG	Proangiogenic matrix
Poly(hydroxyethyl methacrylate)	PHEMA	Drug/protein delivery Stem-cell differentiation Cardiac tissue scaffold[a] Drug-eluting contact lens
Poly(*N*-isopropyl acrylamide)	PNIPAAm	Injectable glucose sensor
Poly(*N*-vinyl-2-pyrrolidone)	PNVP	Spinal tissue scaffold[a]
Poly(vinyl alcohol)	PVA	Cartilage scaffold Drug-eluting sensor coating

Hydrogels for drug delivery systems

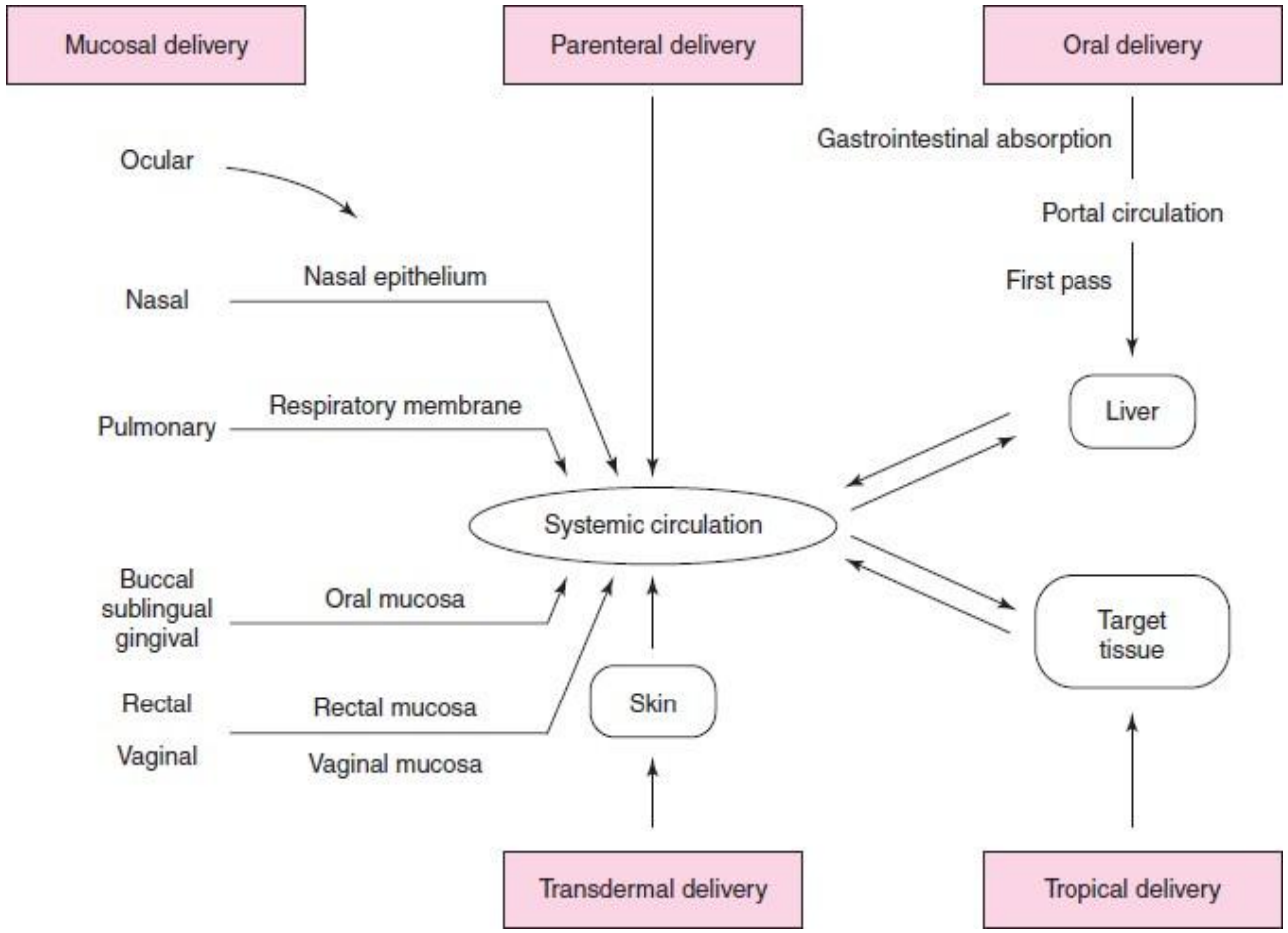


Diffusivity

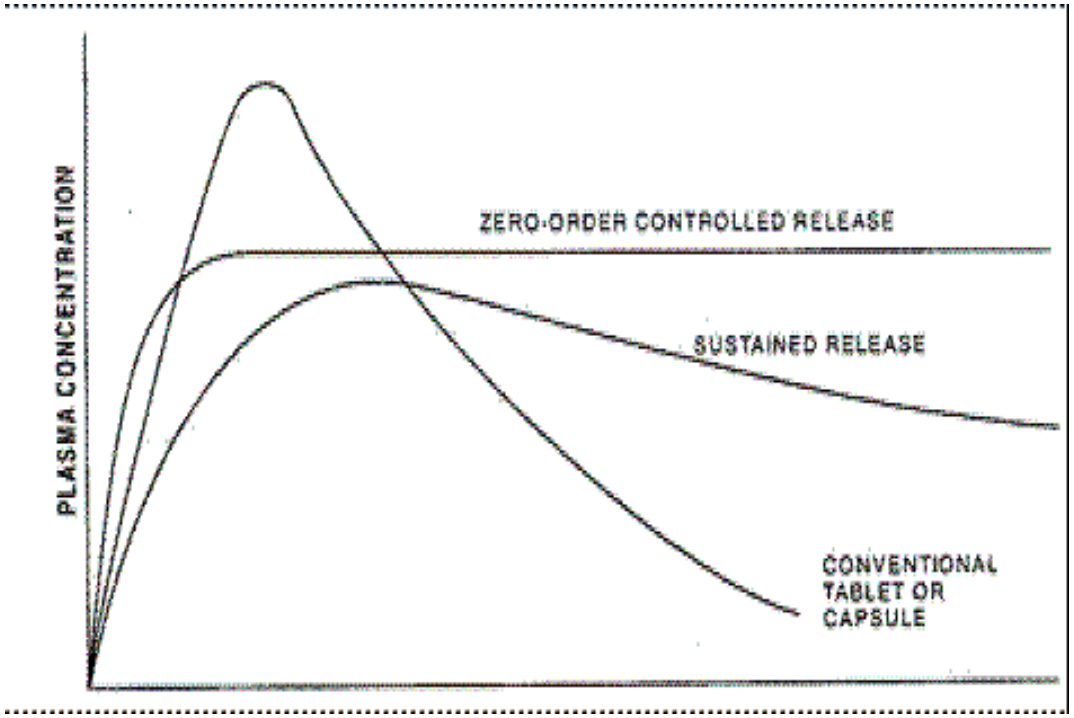
- The ability of a hydrogel to restrict the diffusive movement of a solute, plays a key role in applications like cell encapsulation, chromatography, biosensors, and drug delivery.
- The polymer chains within these networks retard solute movement by reducing the average free volume per molecule available, increasing the hydrodynamic drag experienced, and acting as physical obstructions that increase the solute path length
- Drug release is controlled by diffusion and polymer relaxation

Drug delivery

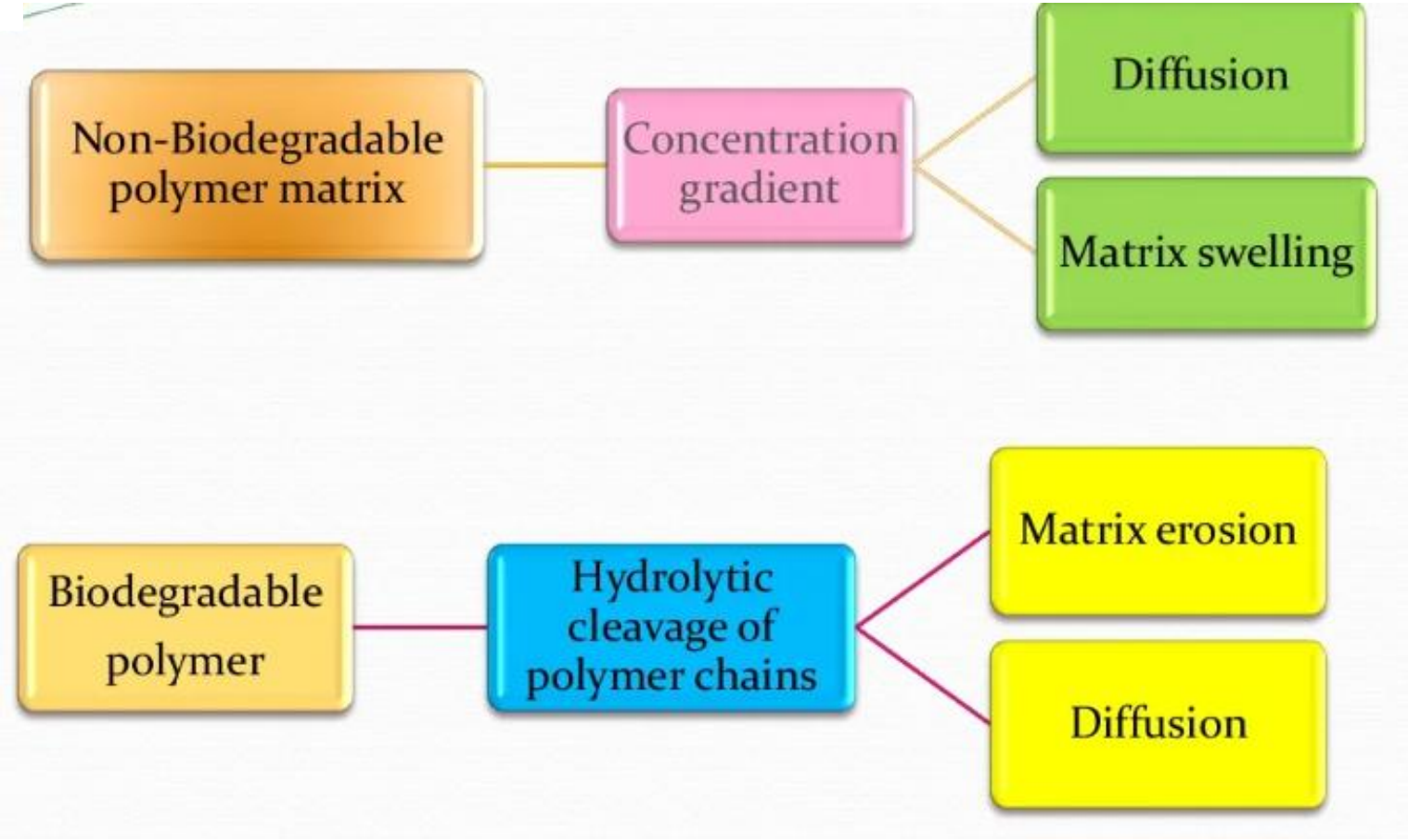
Routes for drug delivery



Requirement for drug delivery system



Drug release mechanism from matrix



FICK'S FIRST LAW

$$J = \frac{dM}{s\ dt}$$

dM = change in mass of material.
S = barrier surface area.
dm/dt = rate of mass transfer.
J = flux.

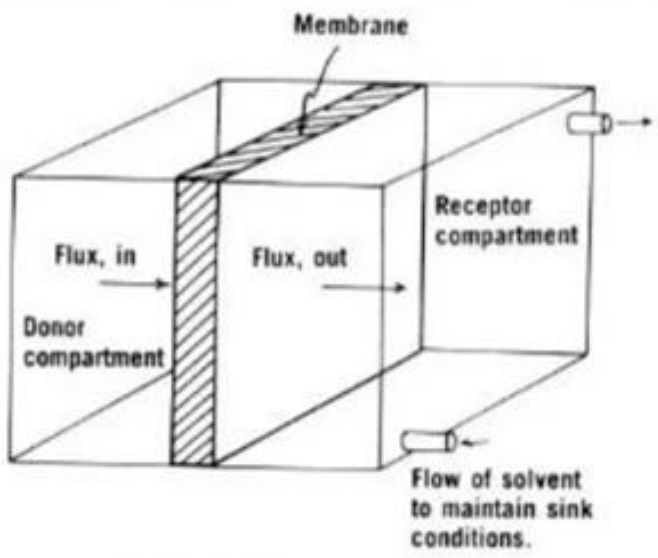


Fig. diffusion cell .donor compartment contains diffusant at conc. c

★ Deals with the mass diffusing across a unit area of barrier in unit time .

FICK'S SECOND LAW

- An equation for mass transport that emphasizes the change in concentration with time at a definite location.
- It states "the change in conc. with time in particular region is proportional to change in conc. gradient at that point in the system."

$$\frac{dC}{dt} = D \frac{d^2c}{dx^2}$$

Zero order equation: Concentration Independent

The equation for zero order release is

$$Q_t = Q_0 + K_0 t$$

Q_0 = initial amount of drug

Q_t = cumulative amount of drug release at time "t". (released occurs rapidly after drug dissolves.)

K_0 = zero order release constant.

t = time in hours

First order equation: Concentration dependent

The first order release equation is-

$$\text{Log } Q_t = \text{Log } Q_0 + Kt/2.303$$

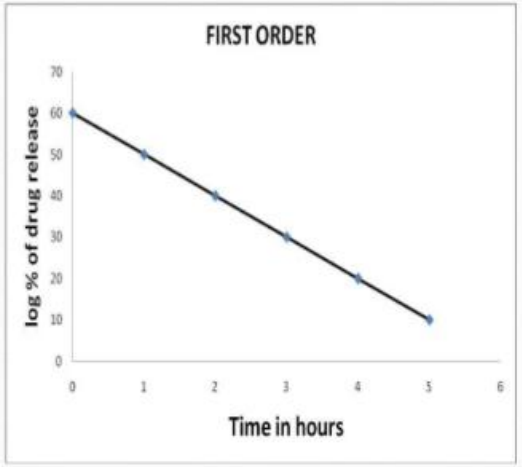
$$\text{or } Q_t = Q_0 e^{-Kt}$$

Q_0 = initial amount of drug.

Q_t = cumulative amount of drug release at time "t".

K = first order release constant.

t = time in hours.



Generalized Korsmeyer Model

Korsmeyer – peppas equation is

$$F = (M_t / M) = K_m t^n$$

F = Fraction of drug released at time 't'

M_t = Amount of drug released at time 't'

M = Total amount of drug in dosage form

K_m = Kinetic constant

n = Diffusion or release exponent

t = Time in hours

n= 0.5

- Diffusion release
- Square root of time depended.
- Fickian release.

n= 1

- Swelling controlled
- Release rate independent of time.
- Zero order/case II .

n= between 0.5-1

- Diffusion and swelling
- Release is time depended.
- First order/anamolus /non-fickian.

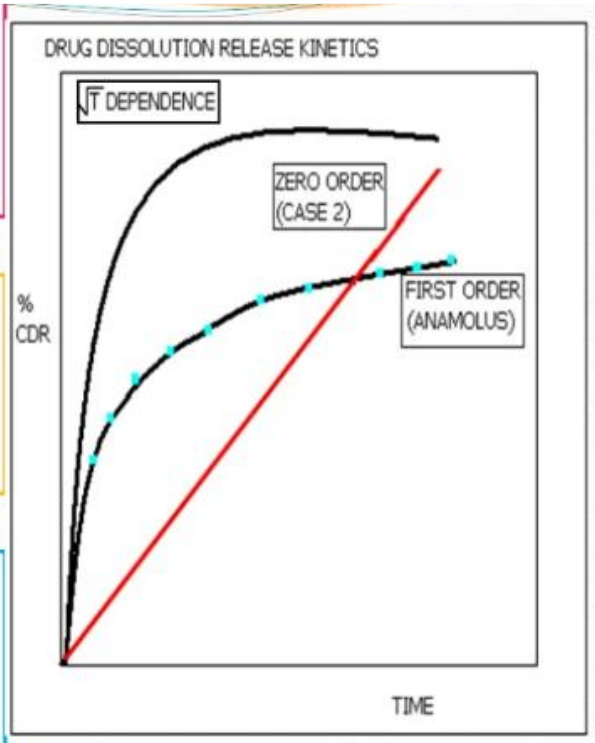
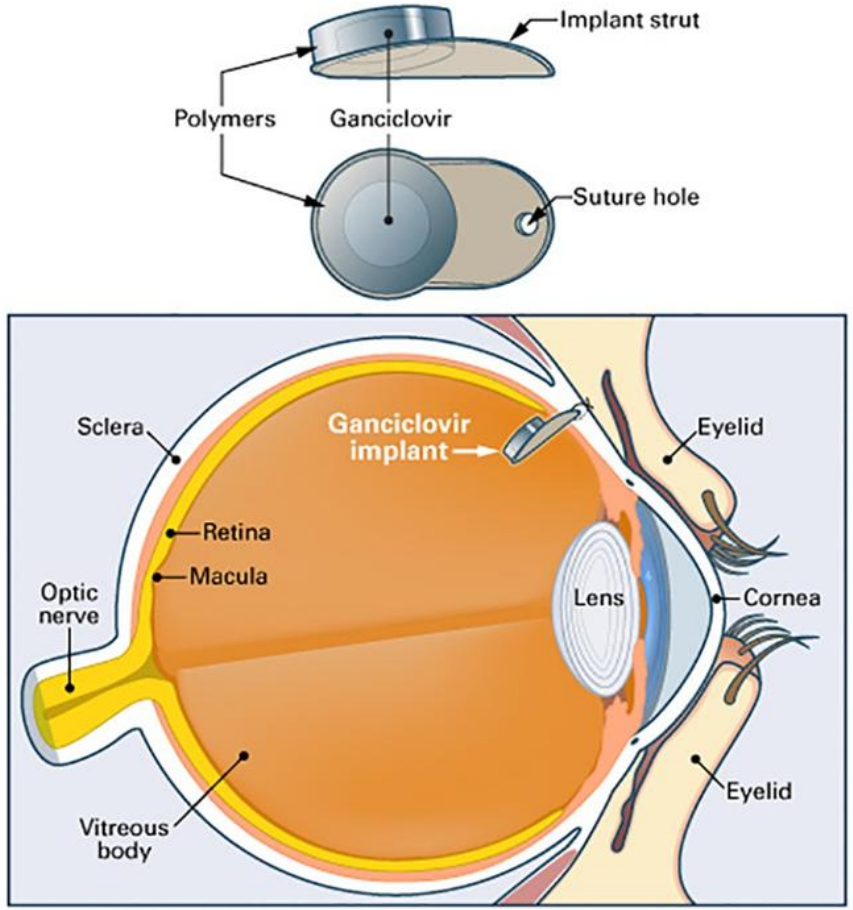
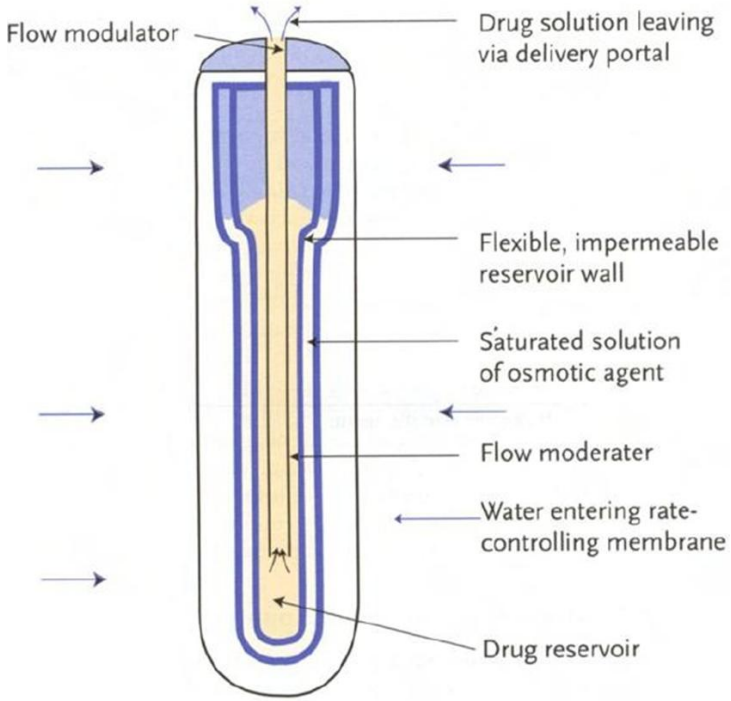


fig:- graph of drug release kinetics



Vitrasert (Ocular drug delivery)

Implantable contraceptive for young adults



Implantable drug delivery osmotic pump

Drug delivery implants

Advantages

- Convenience and compliance

- Potential for rate controlled and bioresponsive release

- Improved drug delivery locally or systemically with minimal interference from biological or metabolic barriers

- Flexibility with choice of material, manufacturing process, drug loading and release

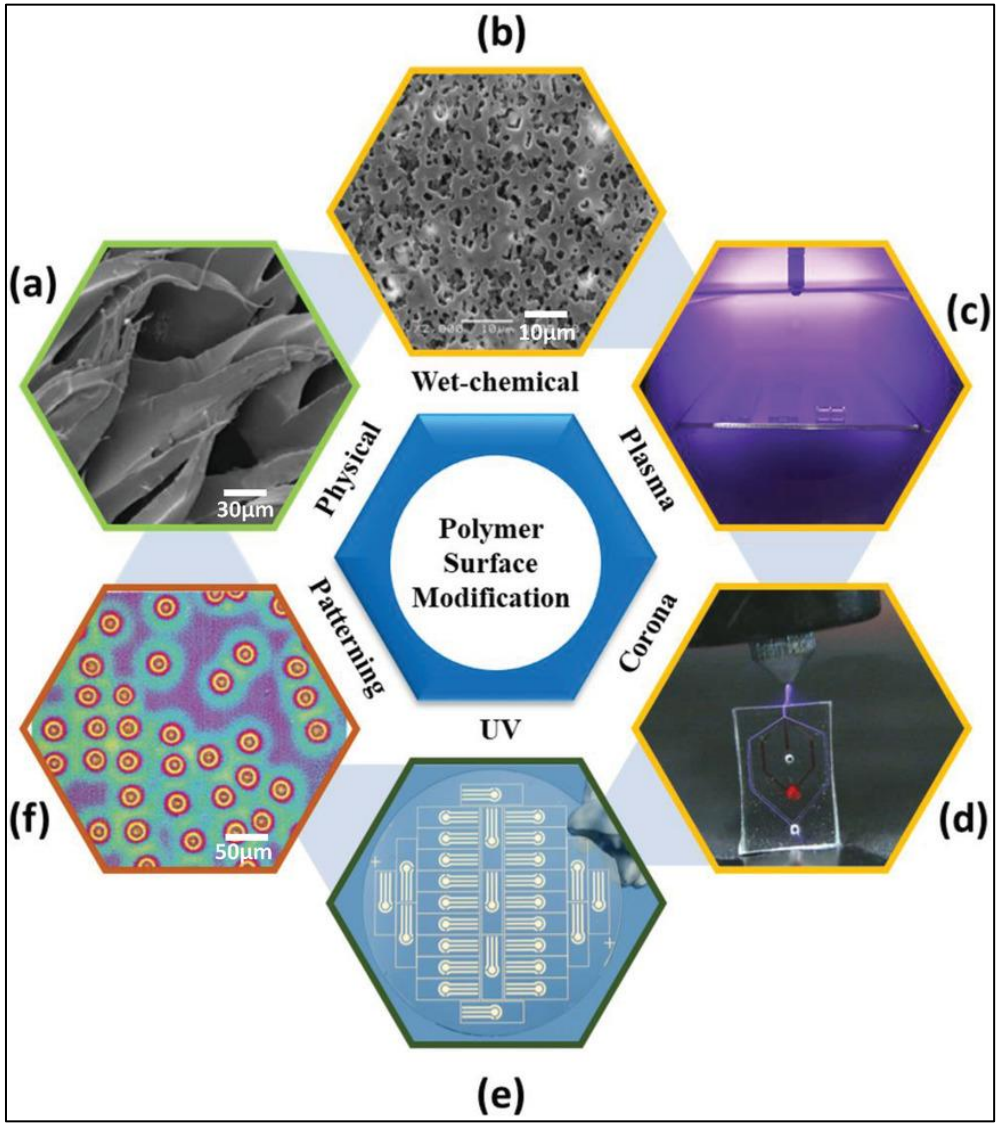
- It is regarded as new drug product and can extend the market exclusivity (through patents) by five years

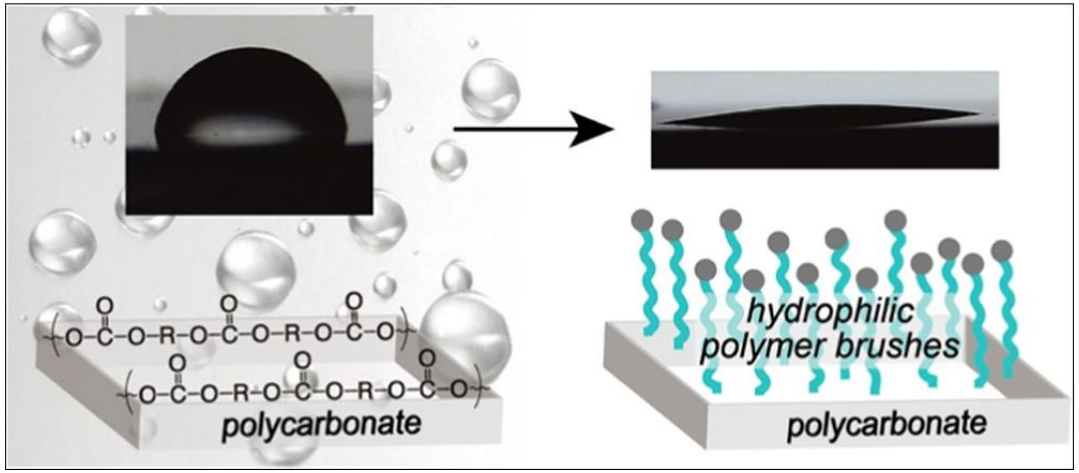
Surface Modification

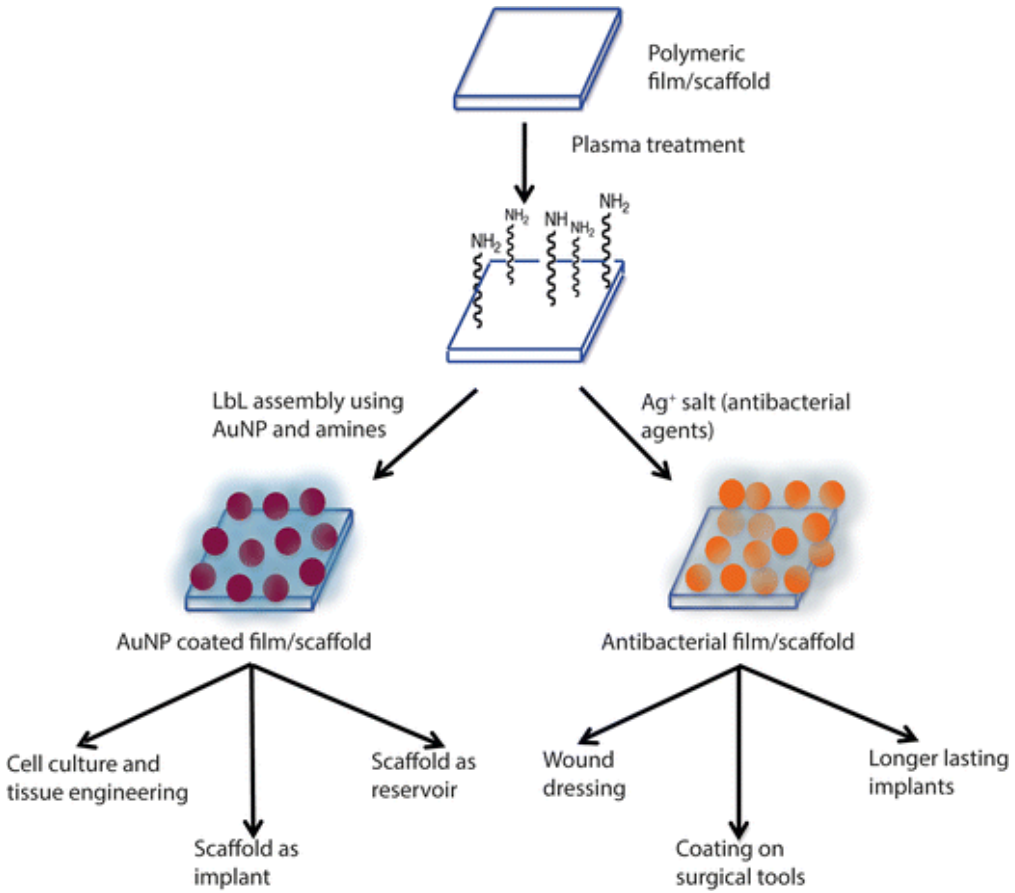
TABLE 3.7 Physical and Chemical Surface Modification Methods for Polymeric Biomaterials

To modify blood compatibility	Octadecyl group attachment to surface
	Silicon containing block copolymer additive
	Plasma fluoropolymer deposition
	Plasma siloxane polymer deposition
	Radiation-grafted hydrogels
	Chemically modified polystyrene for heparin-like activity
To influence cell adhesion and growth	Oxidized polystyrene surface
	Ammonia plasma-treated surface
	Plasma-deposited acetone or methanol film
	Plasma fluoropolymer deposition
To control protein adsorption	Surface with immobilized polyethylenglycol
	Treated ELISA dish surface
	Affinity chromatography particulates
	Surface cross-linked contact lens
To improve lubricity	Plasma treatment
	Radiation-grafted hydrogels
	Interpenetrating polymeric networks
To improve wear resistance and corrosion resistance	Ion implantation
	Diamond deposition
	Anodization
To alter transport properties	Plasma deposition (methane, fluoropolymer, siloxane)
To modify electrical characteristics	Plasma deposition
	Solvent coatings
	Parylene coatings

Surface modifications:





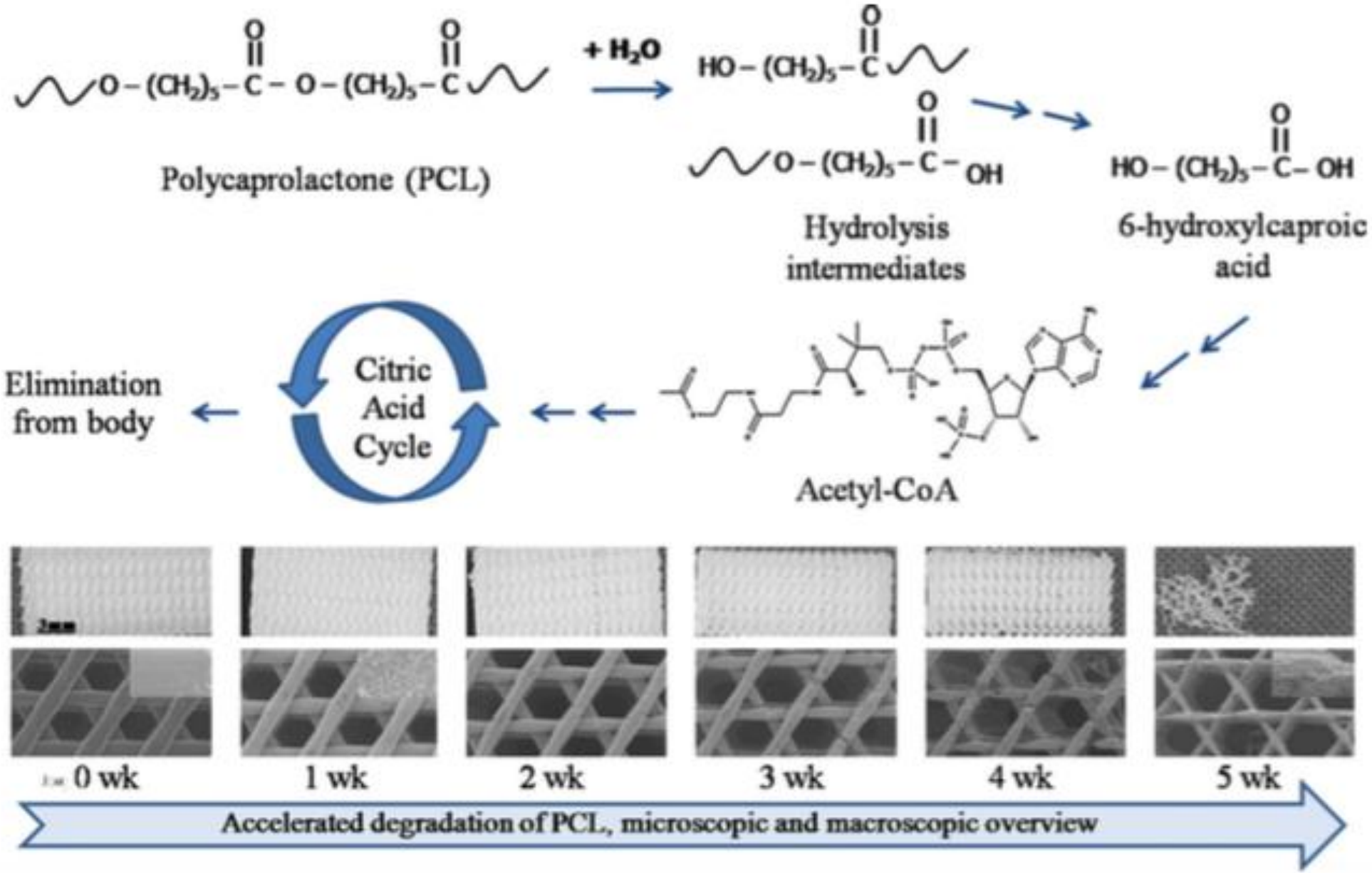


Biodegradation of polymers:

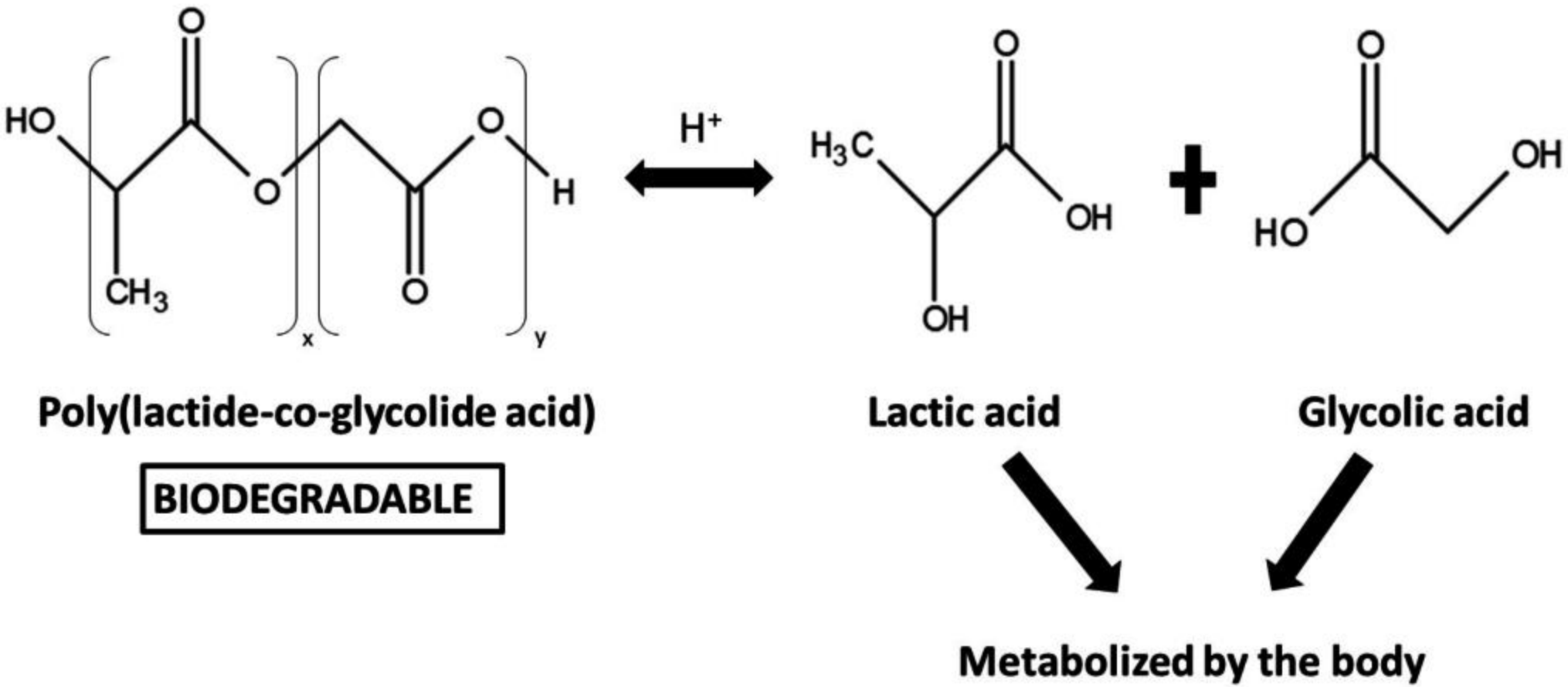
Degradation of polymers in presence of biological system or bio-mimetic systems

- Physical degradation: degradation in methods like photodegradation and hydrolysis
- Biological Degradation:
- Aerobic method: Degradation (presence of oxygen) into water, carbon dioxide, and biomass
- Anaerobic method: Degradation (absence of oxygen) into carbon dioxide, methane and biomass for testing in an in vitro systems, a material is incubated at body temperature (37 degree C) in phosphate buffered saline with enzymes like lysozyme, pepsin or trypsin.

Biodegradation of PCL

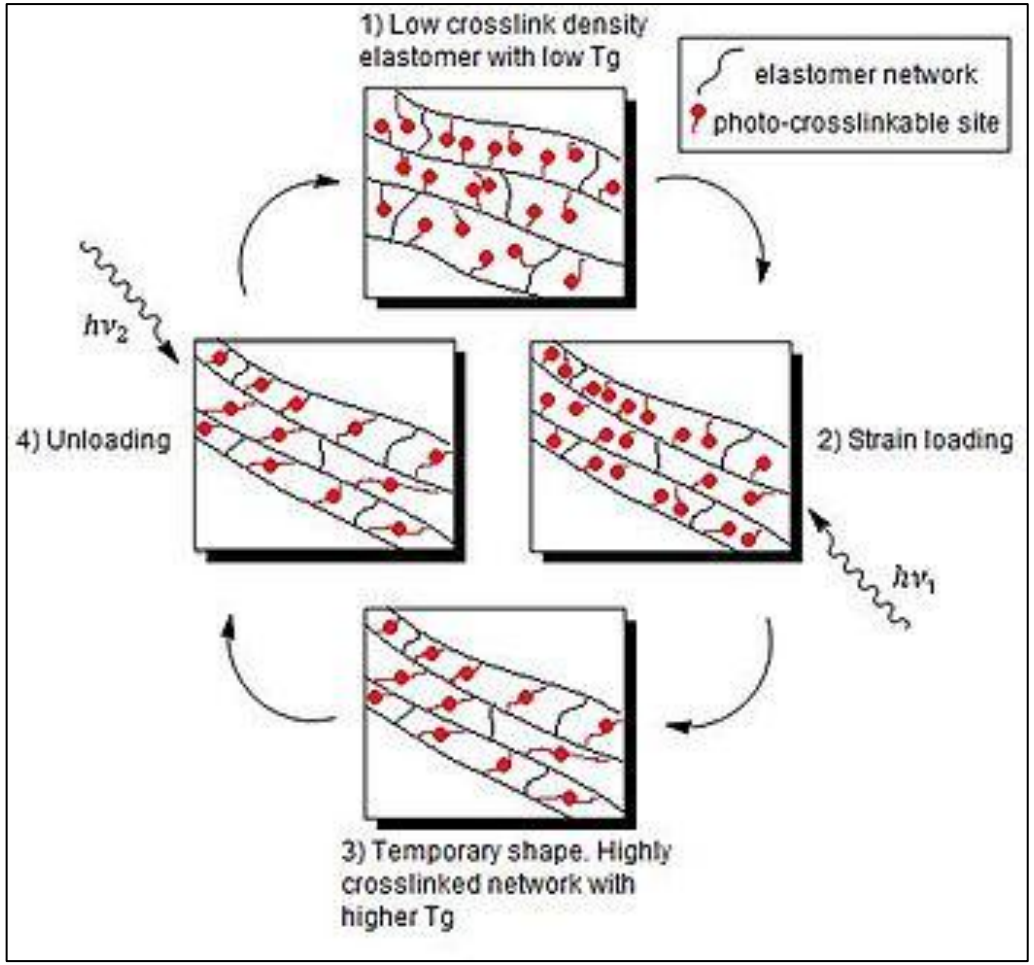


Biodegradation of PLGA



Shape memory polymers: The polymers can recover from deformed shape by physico-chemical triggers

- Thermal trigger: Polyurethane, Polyethylene terephthalate
- Light trigger: Cinnamic acid, Cinnamyldene acetic acid
- Electric trigger: Elastomer-MWCNT/Graphene composite

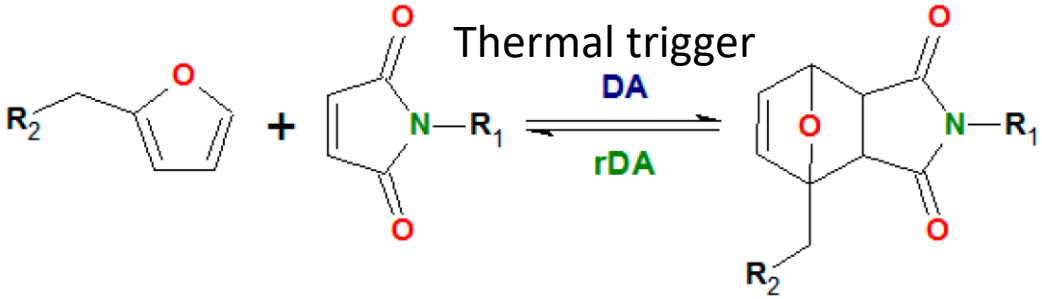


Application: Orthopedic surgical devices, microsurgery tools, ophthalmic surgery tools

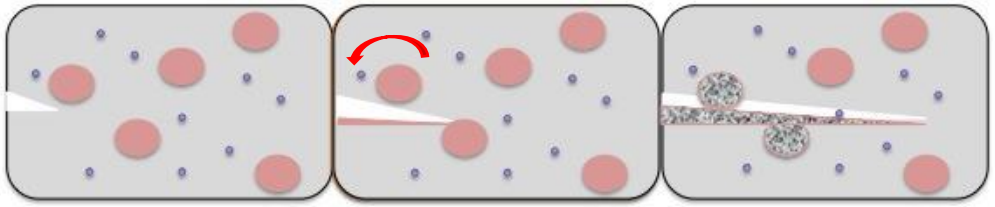
Self healing polymers:

- Polymers can repair up to a limit when damaged mechanically.
- Intrinsic polymer self healing: The polymer chains near the damage helps in healing of the polymer chain:

  Example: Cross linking of linear thermo plastic chain between pendant groups

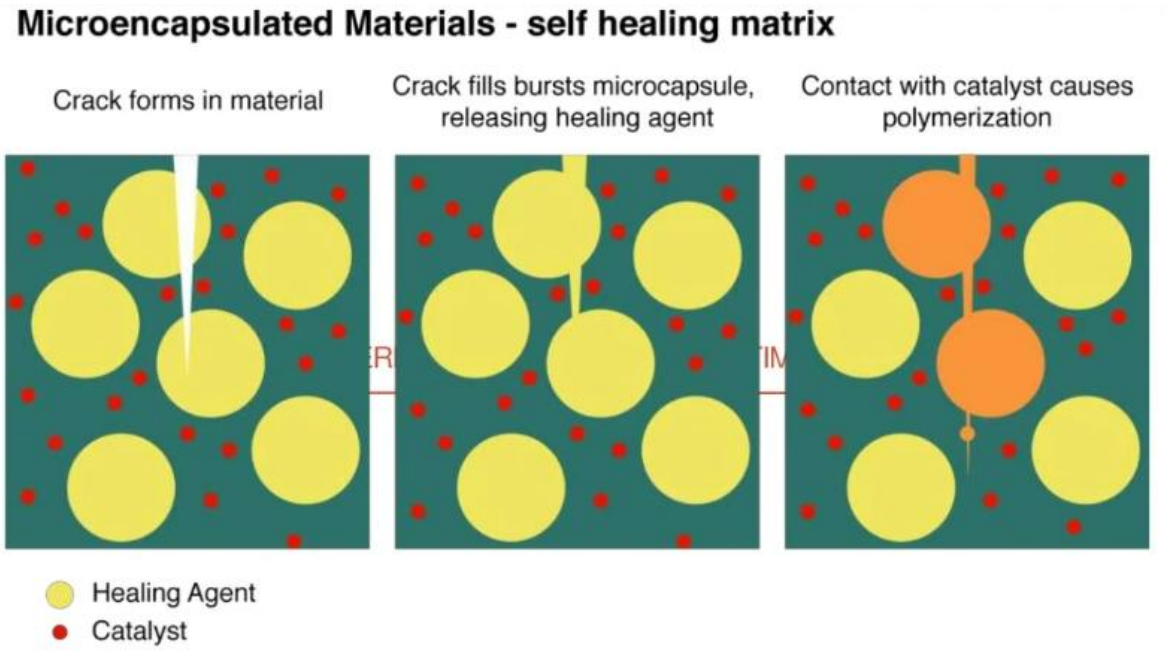


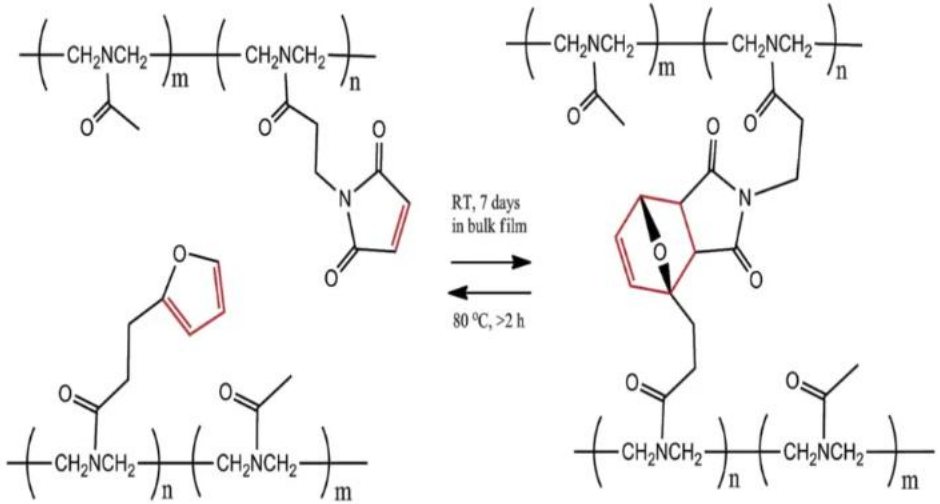
Extrinsic polymer self healing: The healing chemistries are separated from the surrounding polymer in microcapsules or vascular networks which, after material damage/cracking, release their content into the crack plane, reacting and allowing the restoration of material functionalities



Microcapsule healing

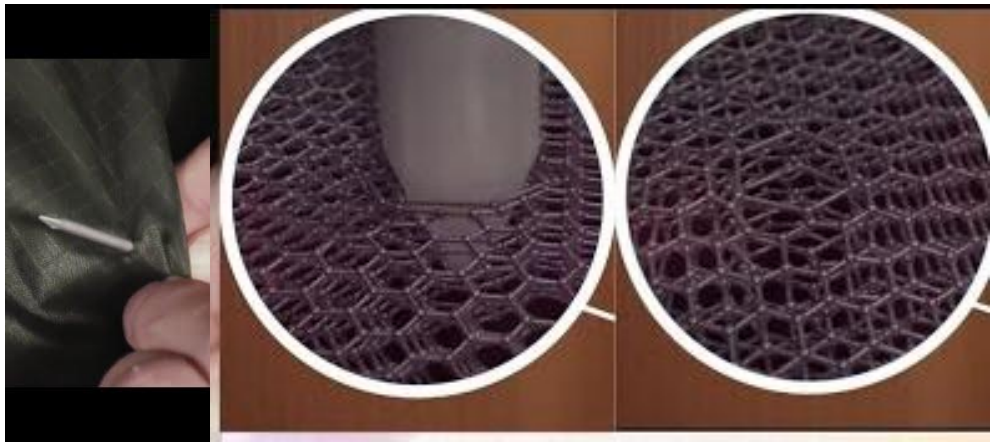
Engineering of self healing polymers

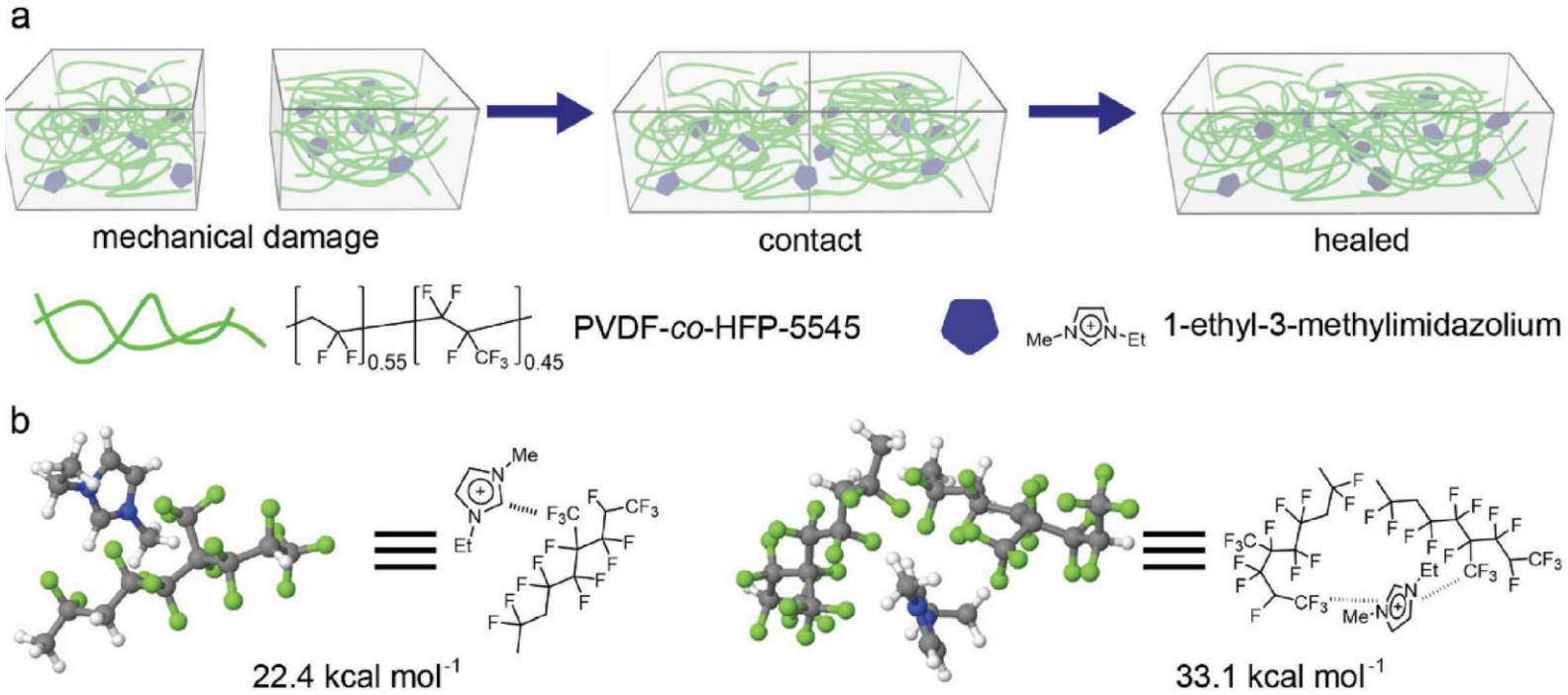




Self-healing materials are artificial or synthetically-created substances that have the built-in ability to automatically repair damage to themselves

Wolverine material

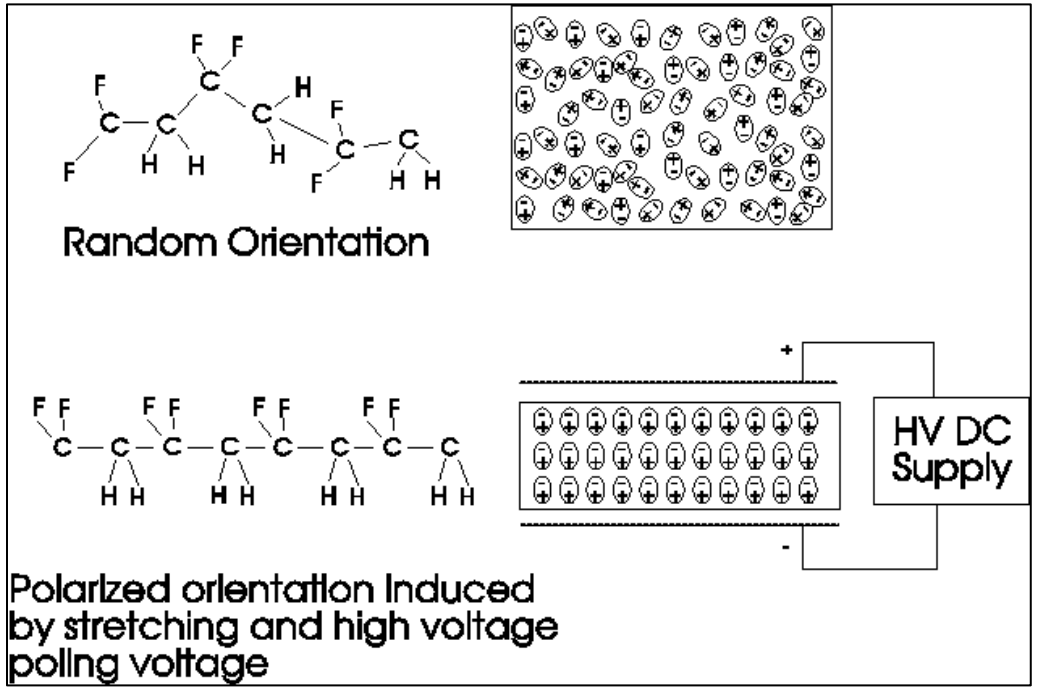




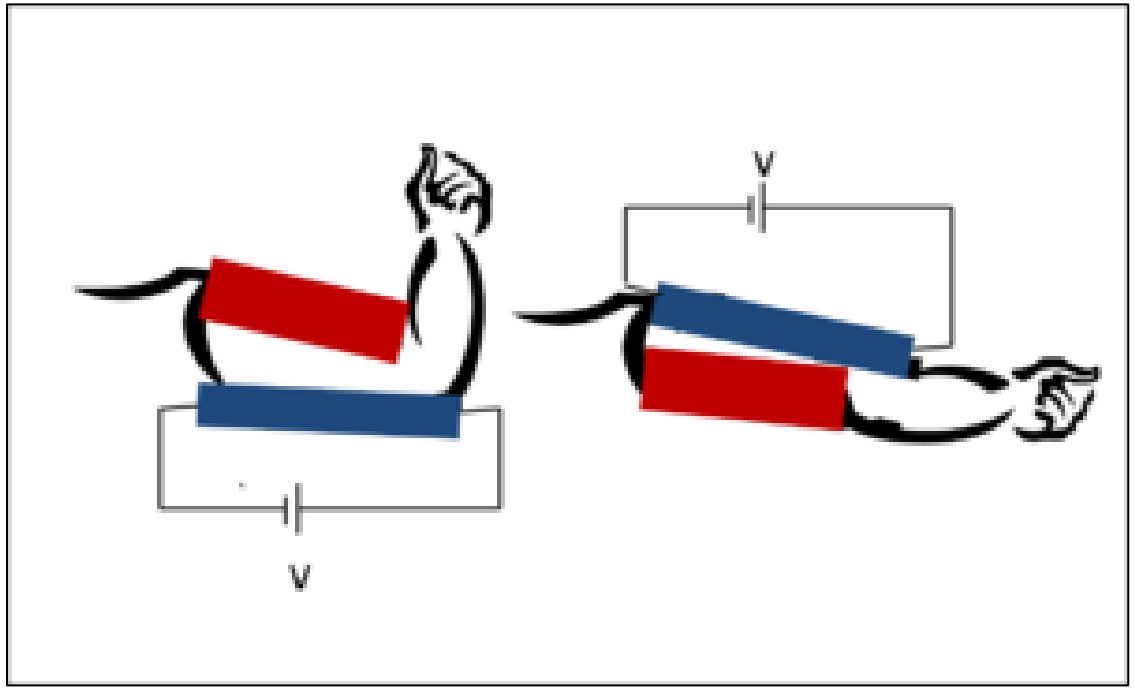
Electroactive Polymer:

Used to develop biomedical actuators:

Example : PVDF (Polyvinylidene fluoride) Piezo electric polymer, Ionic polymer-metal composite

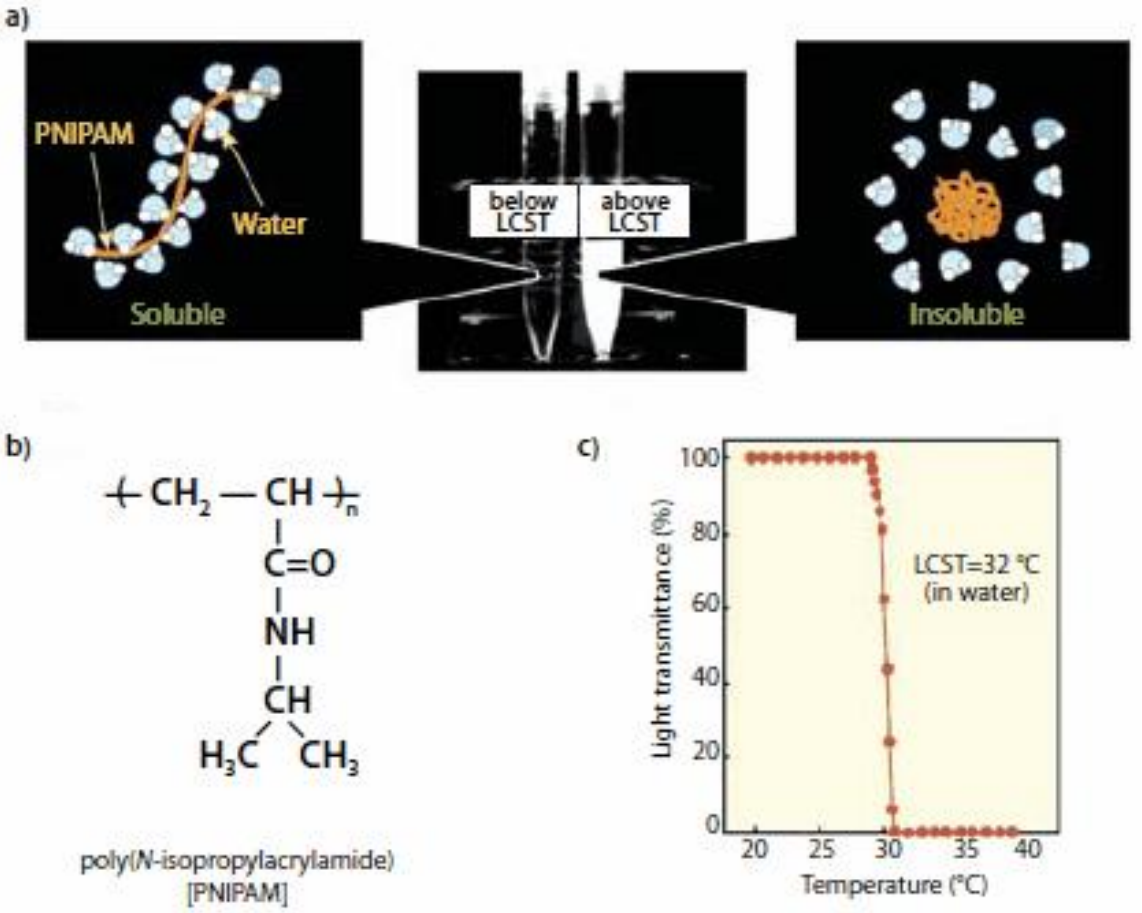


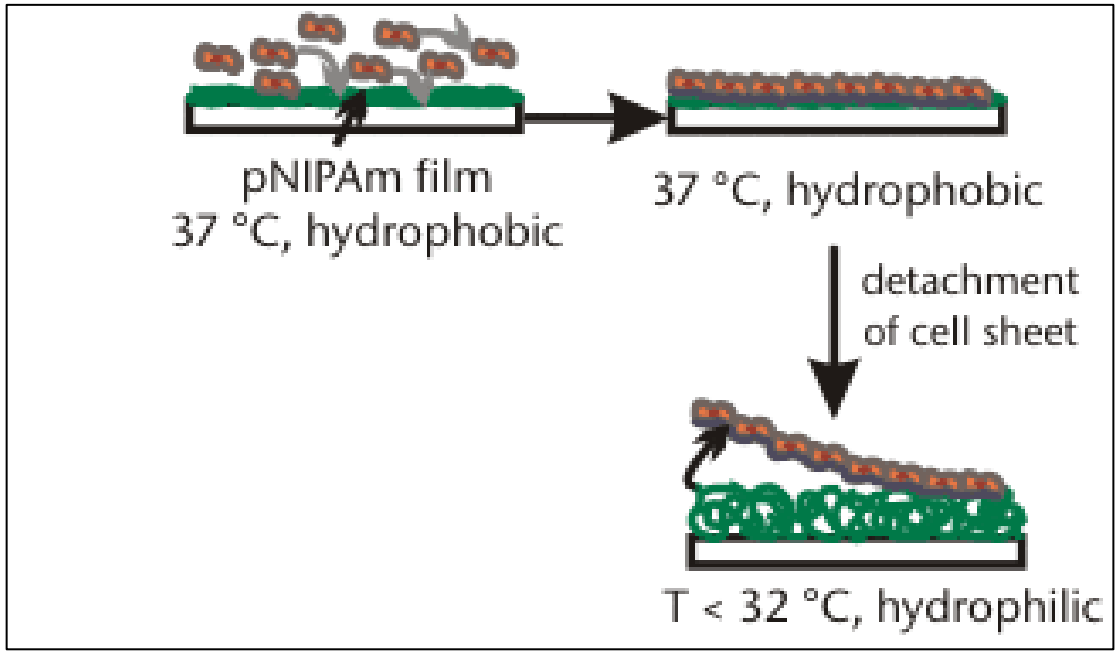
Polarizing of PVDF membrane



Artificial muscle prepared by electroactive polymer

Thermo-active polymers



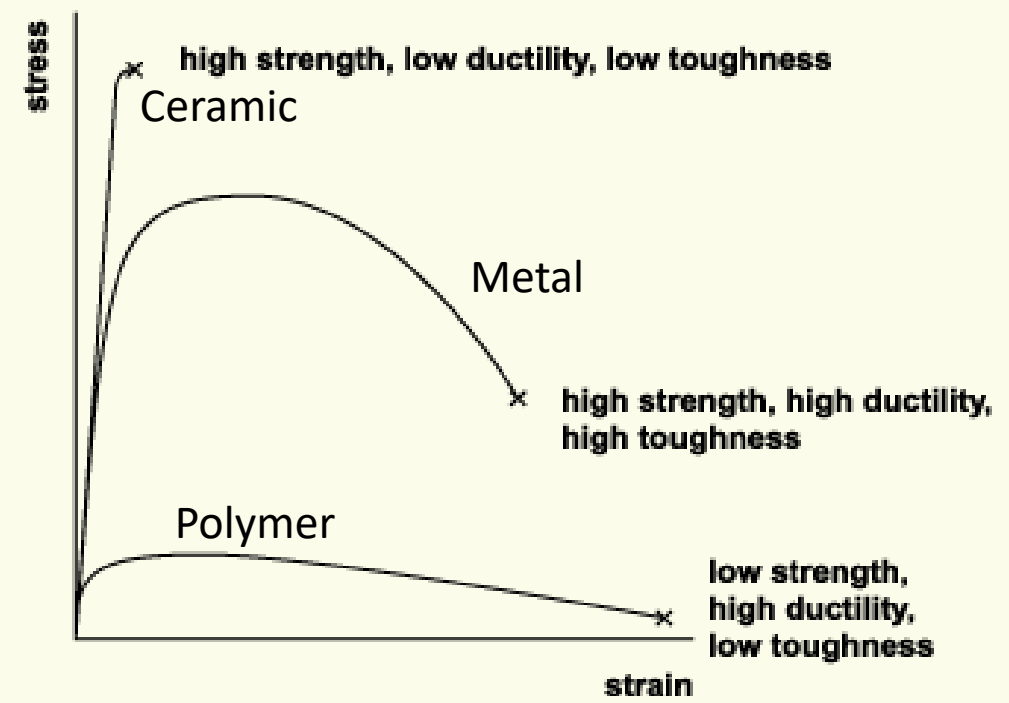


Cell sheet engineering by PNIPAM thermal switching

Poly(N-isopropylacrylamide): PNIPAM (Lower Critical Solution Temperature, LCST beyond which suddenly the hydrogel loses more than 90% of the water content

Ceramics (evolved from Greek word “keramikos”..Pottery)

- Traditionally metal or non-metallic oxide materials
- Polycrystalline material developed in very high temperature
- Mechanically brittle, high compressive strength, poor against bending
- Structural rigidity is the result of presence of both covalent and ionic bonds
- Aesthetic looking
- Poor conductor of heat and electricity



Teacup

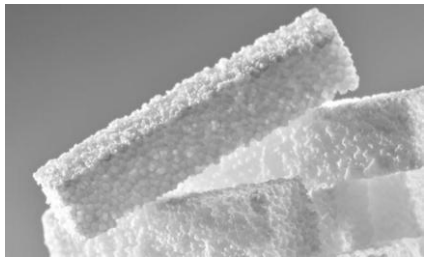
Porcelain Vase

Bioceramics in medical engineering:

- Bioinert: Neither cross react with body nor form an active tissue bonding
  Example: Alumina, Zirconia
- Bioactive: Forms active bonding with tissue but do not completely degrade over time.
  Example: Sintered Hydroxy apatite, β-tricalcium phosphate, Bio-glass
- Bioresorbable: Not only forms active bonding to tissues but degrades over time
  Example: Ortho-calcium-phosphate, Hydroxyapatite, Calcium Sulfate

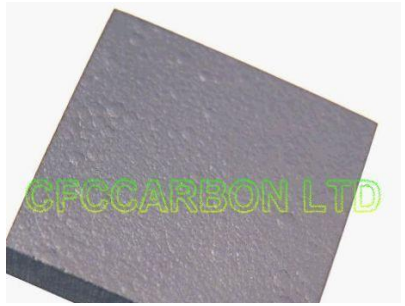
Alumina

Zirconium



Calcium Phosphate

Silica



Pyrolytic Carbon

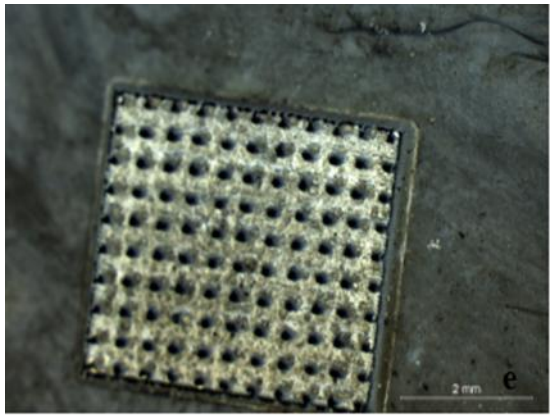
Hydroxyapatite

Bioinert Ceramic Materials: Alumina and Zirconia:

Alumina total dental implant

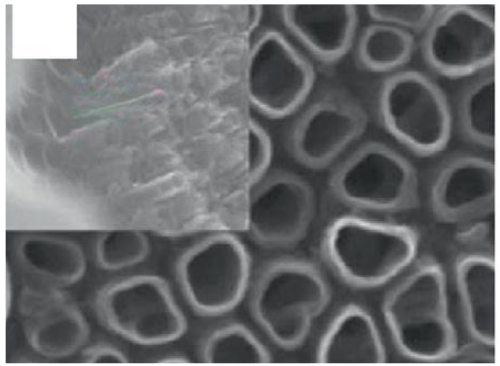
Zirconia dental screw and root

Zirconia femoral head of hip implant

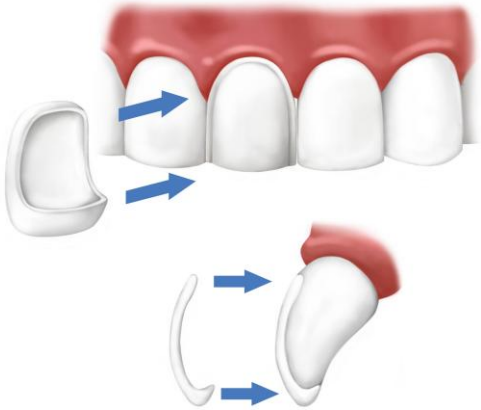


Alumina microfluidic chamber

Alumina based porous bio-eye



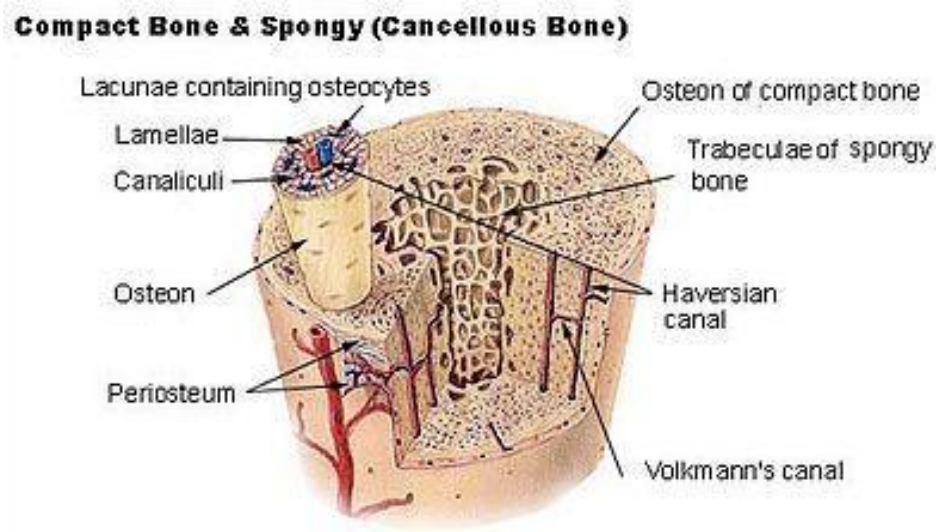
Titanium Oxide nanotubes for implant surface activation

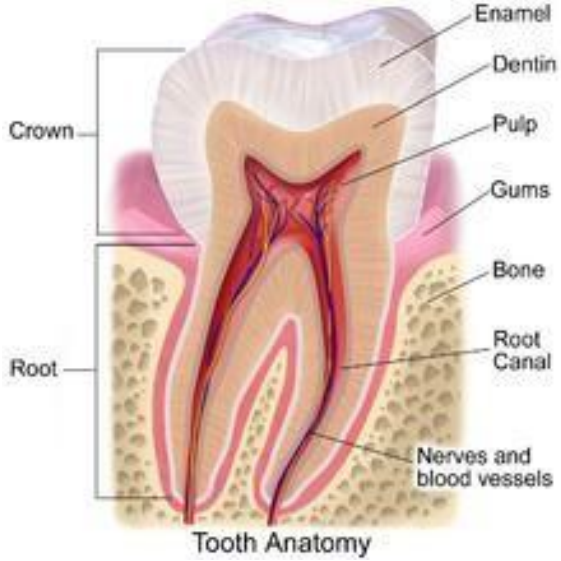


Porcelain cap

Veneering or formation of small shell

Major application of bioactive ceramics



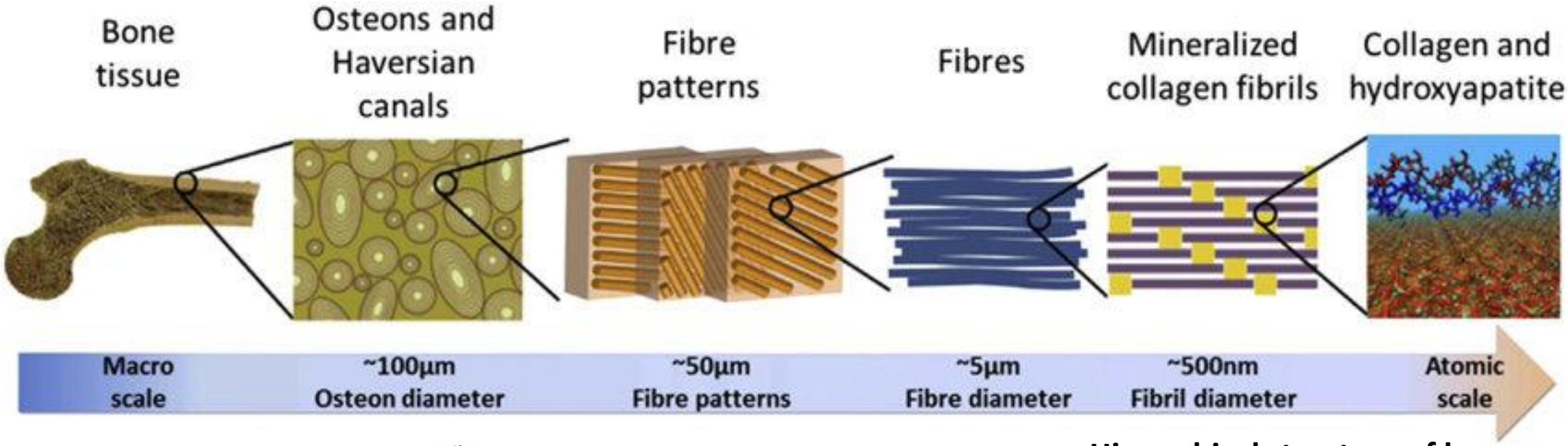


Bones and teeth are ceramic-polymer composites

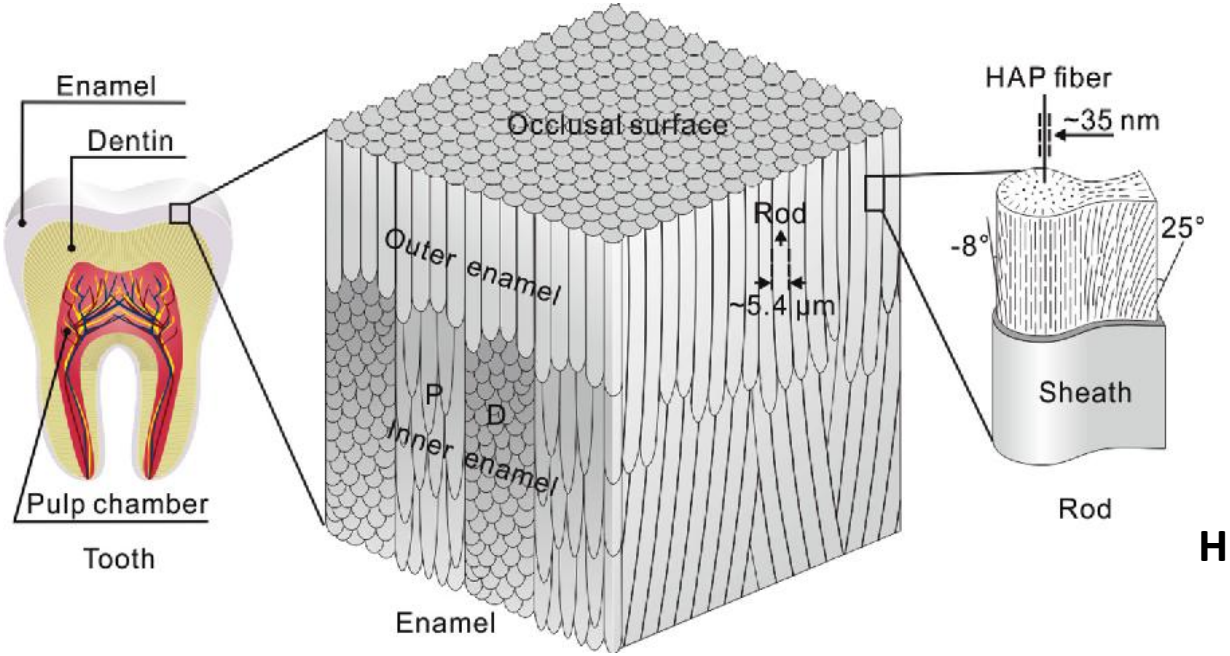
Bone tissues: Osteoblast cells releasing a matrix of type I collagen and calcium, magnesium, and phosphate ion, which combine chemically in the matrix in the form of hydroxyapatite and known as <u>bone mineral</u>

Teeth: Primary mineral in <u>enamel</u> is hydroxyapatite and <u>dentin</u> contains organic matrix of collagenous proteins

Design of Bone and Teeth from macro to micro to nano regime



Hierarchical structure of bone



CaP is the major material for bone and teeth mineralization

Hierarchical structure of tooth enamel

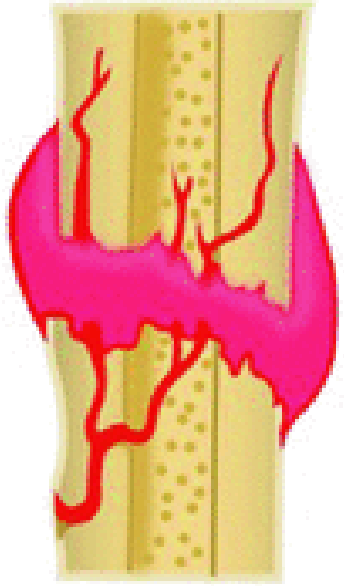
Three major terms related to bone regeneration:

- Osteoconductive: Bone growth starts on a surface (like bioactive implants) via active bond formation. If the materials is biodegradable and allows angiogenesis in long run can lead to osteogenesis or synthesis of new bone

- Osteoinduction: Osteoinduction implies the recruitment of immature cells (precursor/progenitor/stem cells) and the stimulation of these cells to develop into preosteoblasts. In a bone healing situation such as a fracture, most of the bone healing is dependent on osteoinduction.

- Osseointegration: Osseointegration is the stable anchorage of an implant achieved by direct bone-to-implant contact. It is not bone growth on the surface but bone growth around the implant (no chemical bonding but formation of mechanical anchorage)

Stages of Bone Fracture Healing

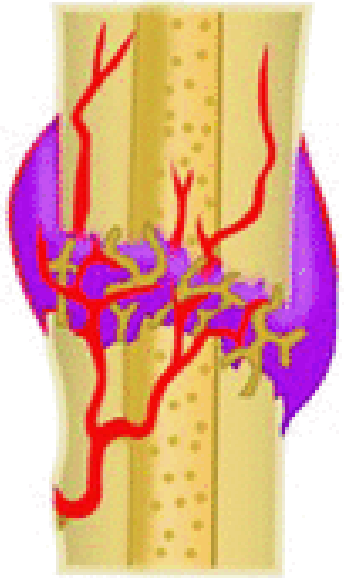


Haematoma

- Activation of coagulation cascade
- Changes of local environment
- Inflammatory cells and molecules released

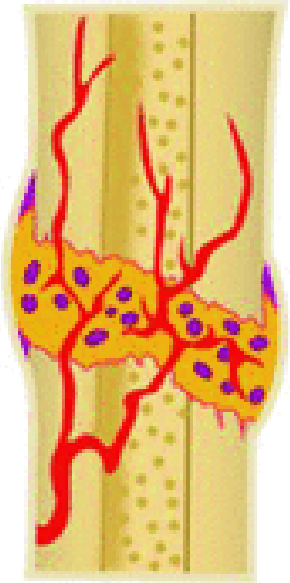
Inflammation

- Recruitment and activation of inflammatory and osteoprogenitor cells
- Clearance of necrotic tissues



Granulation tissue

- Active proliferation of osteoprogenitor cells
- Angiogenesis
- Extracellular matrix production

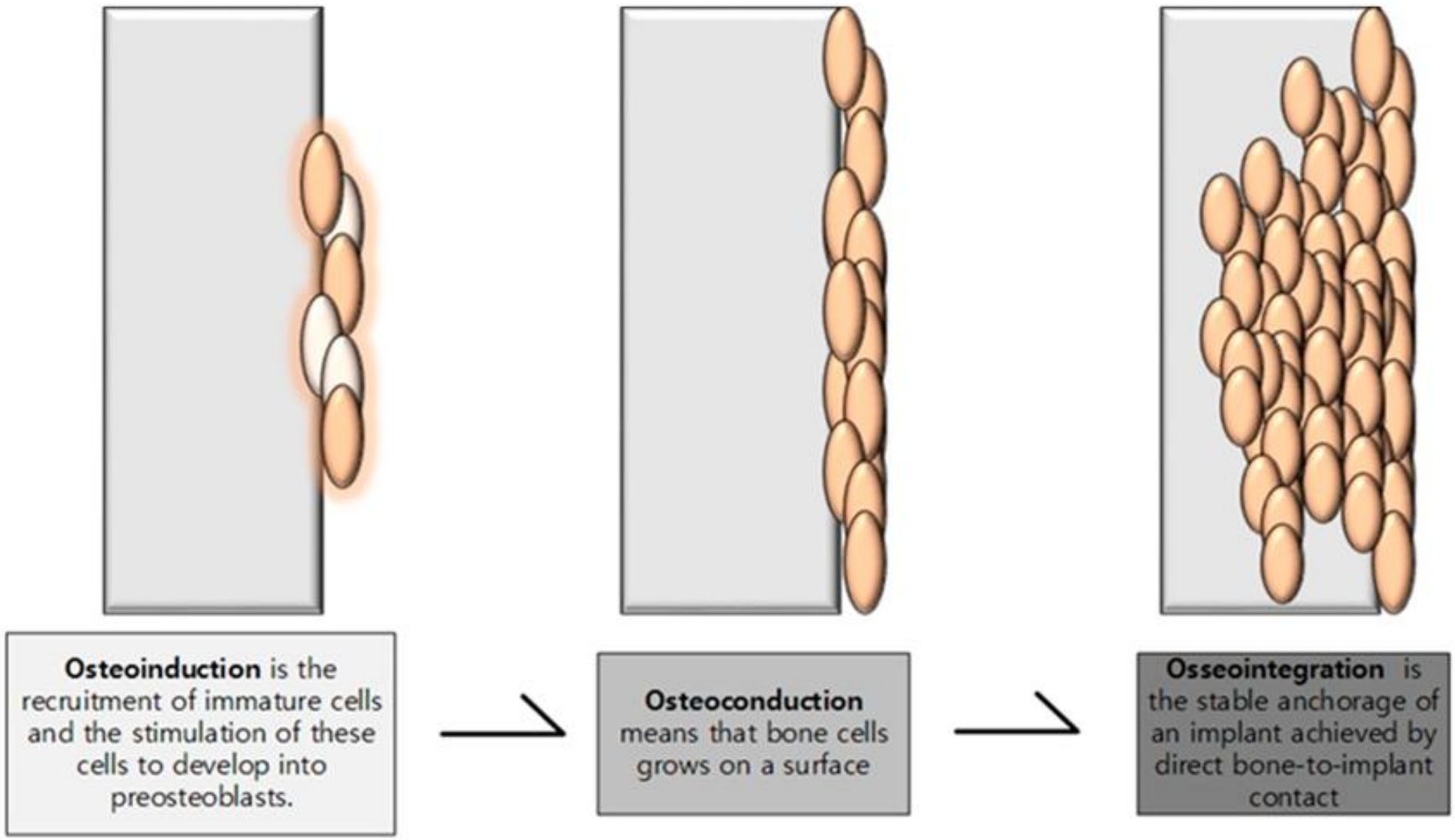


Callus formation

- Soft and Hard
- Differentiation of MSCs according to the mechanical environment
- Initial stabilization of fracture, then replaced by calcified tissue

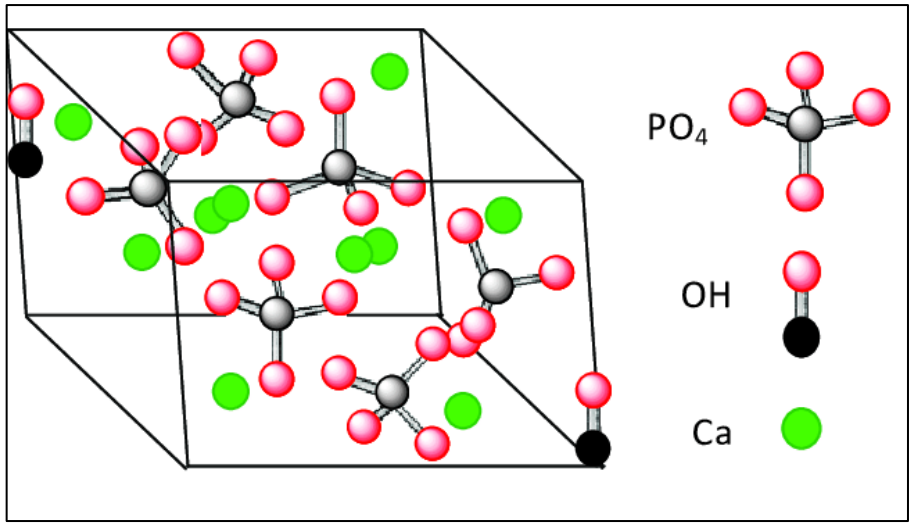
Remodeling

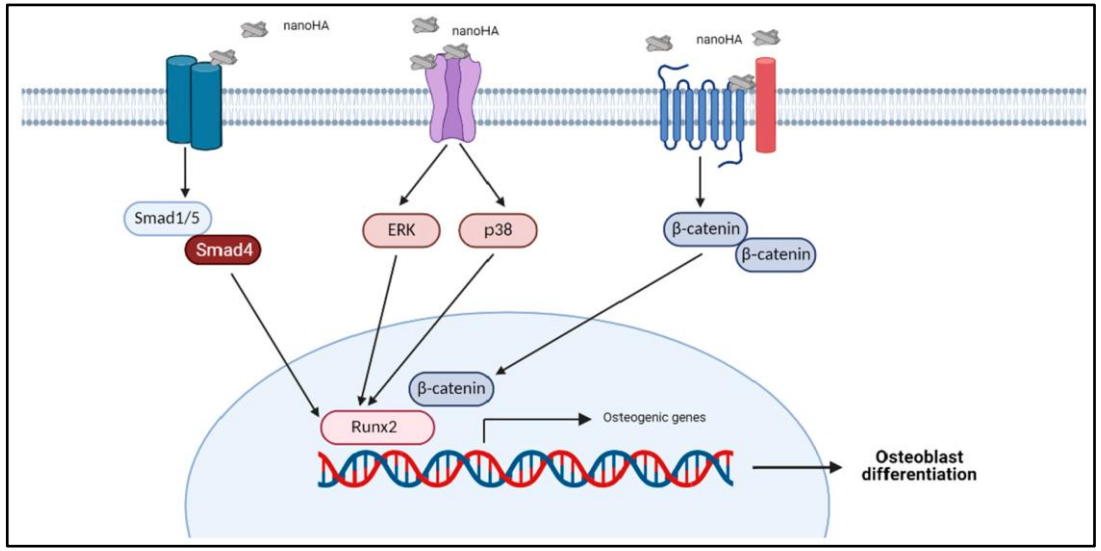
- Long process (years)
- Resorption of remaining cartilage
- Restoration of Haversian system
- No scar formed

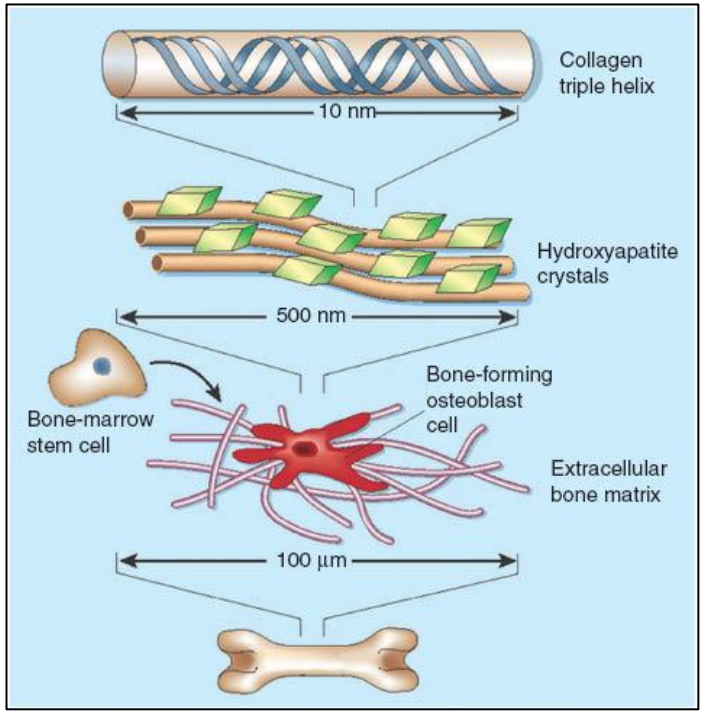

Osteoinduction is the recruitment of immature cells and the stimulation of these cells to develop into preosteoblasts.
Osteoconduction means that bone cells grows on a surface
Osseointegration is the stable anchorage of an implant achieved by direct bone-to-implant contact

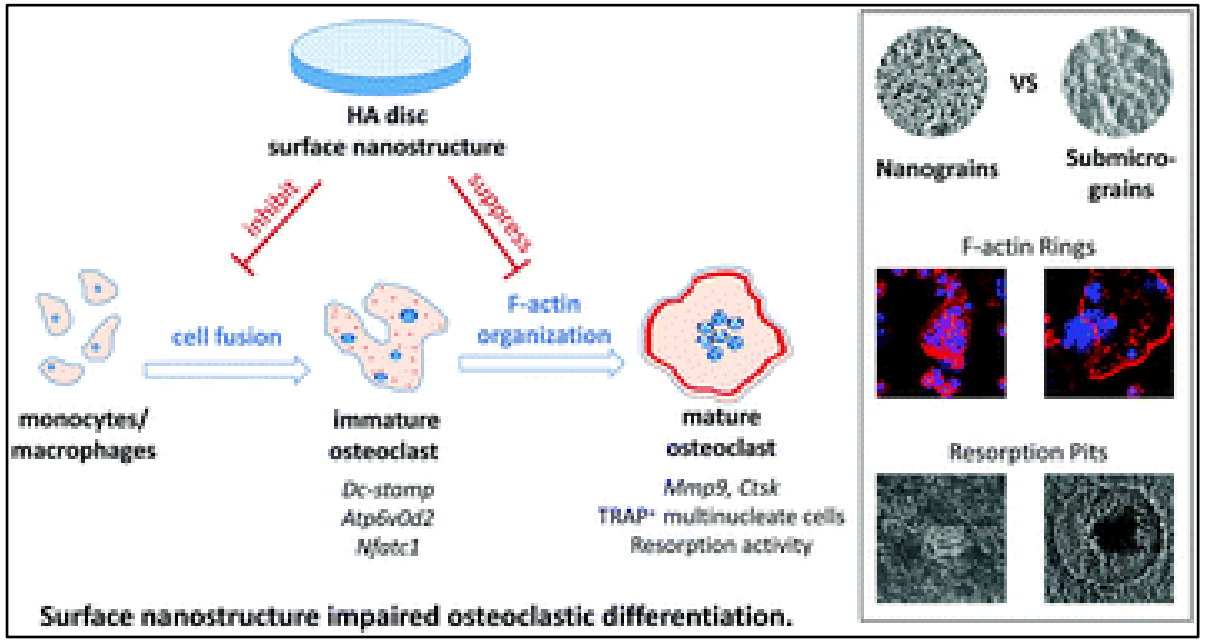
Bioactive Ceramic Materials

HA nanostructures and osteogenic differentiation

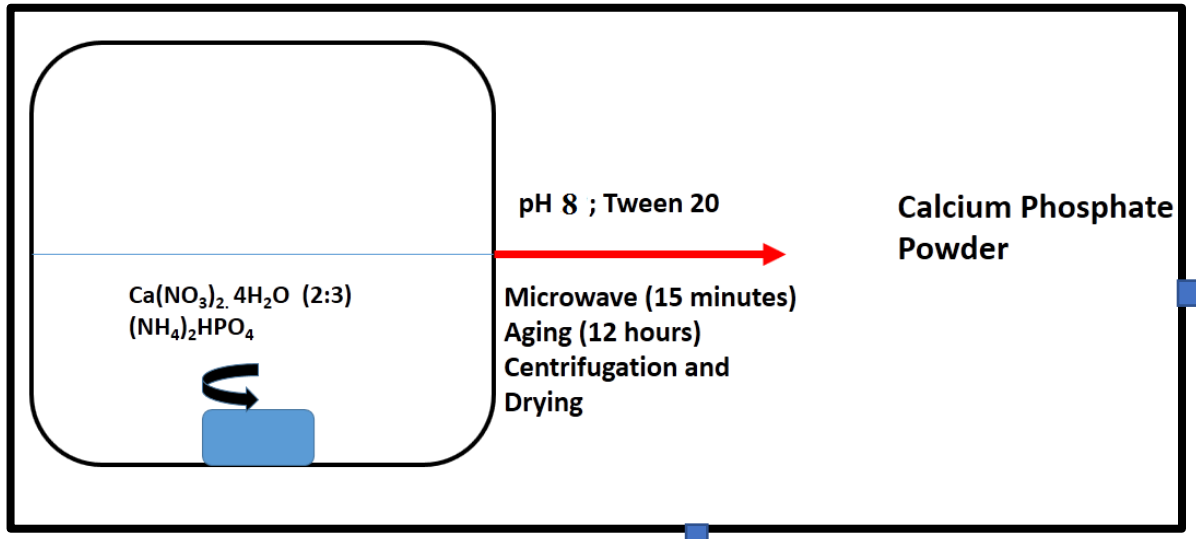


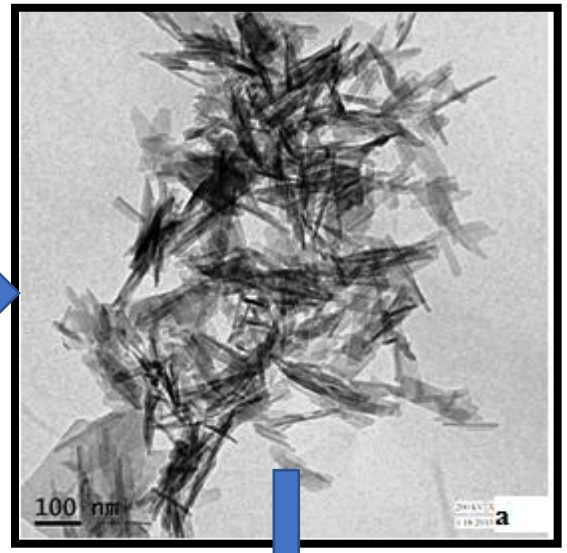






Bioactive Ceramics: CaP, Hydroxyapatite, β-tricalcium phosphate, Bioglass

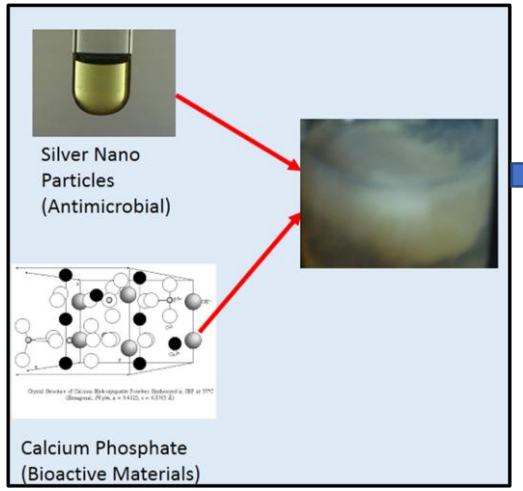


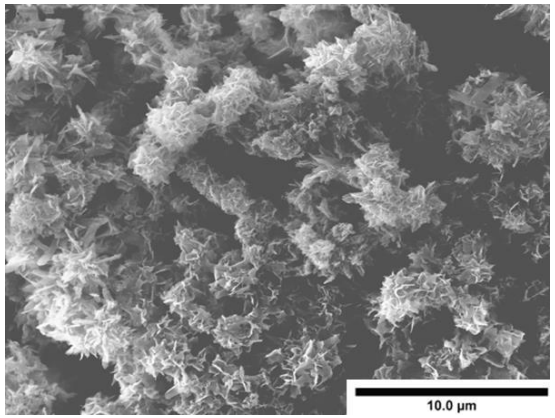


TEM of nanopowder

Solution phase nano particle doping

Foaming and gel casting





Electrodeposition Antibacterial coating

Porous CaP bone graft